NSC GRADE 12 · MATHEMATICS

Paper 1: every-way-it's-asked mock

One mock paper in the real NSC question order. Under each question number you get every way that question was asked in the November papers 2016–2025.

How to use it

- The real paper is 150 marks in 3 hours: algebra, equations and inequalities (about 25), patterns, sequences and series (about 25), functions and graphs (about 35), finance, growth and decay (about 15), differential calculus (about 35) and probability (about 15). This mock keeps that order. The real papers number the questions slightly differently each year (10 to 12 questions); the topics always come in this order.
- Each question here holds ALL the variants, so it is much longer than in the real paper. Do one question at a time; the real exam picks a handful of these.
- Under every sub-question, **Asked:** years shows which November papers asked it that way.
- Use the official information sheet (file 4) as in the exam. Show all calculations, diagrams and graphs you used. Round off to TWO decimal places unless told otherwise. Diagrams are not necessarily drawn to scale.
- Scenarios and numbers are new (not copied), built on the real question patterns. Every memo number was calculated, not guessed.

Q	Topic	Marks here
1	Algebra, Equations And Inequalities	149
2	Quadratic Patterns	75
3	Arithmetic And Geometric Sequences And Series	154
4	Functions: Hyperbola	47
5	Functions: Exponential, Logarithmic And Inverse	100
6	Functions: Parabola, Straight Line And Mixed Graphs	167
7	Finance, Growth And Decay	133
8	Differential Calculus: First Principles And Rules	89
9	Differential Calculus: Cubic Graphs And Tangents	146
10	Calculus: Optimisation And Rates Of Change	74
11	Probability	88
12	Counting Principle	65
	Total in this mock	**1287**

The official past papers and memos for 2016–2025 are in the folder next to this file, if you want to see the originals.

QUESTION 1: ALGEBRA, EQUATIONS AND INEQUALITIES [149 marks of variants]

Every year, 22 to 27 marks, usually in this order: a quadratic equation by factors, a quadratic equation by the formula (two decimals), a quadratic inequality, a surd equation with a rejected root, and a non-linear pair of simultaneous equations (5 or 6 marks). Usually one more of: an exponential equation (common factor, or a quadratic in 2^x), an exponent-law or number-pattern problem (3 to 4 marks, no calculator), or a proof that links the roots with a sequence. Sometimes: a function $f(x)$ with $f(x) < 0$, $f'(x) \geq 0$ or a parameter p for the nature of the roots. Always check surd answers in the original equation.

Solve for x in each of the following (answers correct to TWO decimal places where indicated):

1.1.1 $(2x - 8)(x + 5) = 0$ (2)

Asked: 2016, 2022, 2024, 2025

1.1.2 $x^2 - 8x = 0$ (2)

Asked: 2020

1.1.3 $x^2 + 3x - 28 = 0$ (3)

Asked: 2017, 2018, 2019, 2021, 2023

1.1.4 $3x^2 - 7x + 1 = 0$ (correct to TWO decimal places) (3)

Asked: 2016, 2018, 2019, 2020, 2021, 2022, 2017 (4 marks)

1.1.5 $4x^2 + 1 = 7x$ (correct to TWO decimal places) (4)

Asked: 2023, 2024, 2025

1.1.6 $x^2 - x - 12 > 0$ (3)

Asked: 2018, 2024 (4 marks)

1.1.7 $x^2 - 56 > x$ (4)

Asked: 2022, 2023

1.1.8 $x^2 + 7x \leq -10$ (4)

Asked: 2021

1.1.9 $(2 - x)(x + 5) < 0$ (3)

Asked: 2020

1.1.10 $9x^2 - 4 < 0$ (3)

Asked: 2019

1.1.11 $6x^2 > 9x$ (4)

Asked: 2025

1.1.12 $\sqrt{x + 10} = x - 2$ (5)

Asked: 2020, 2021 (4 marks), 2023 (4 marks)

1.1.13 $\sqrt{2x - 6} + 3 = x$ (5)

Asked: 2016, 2024 (4 marks, one answer rejected)

1.1.14 $\sqrt{x^2 - 12} = 2\sqrt{x}$ (4)

Asked: 2017

1.1.15 $2\sqrt{x} = x - 3$ (4)

Asked: 2018, 2022 (both answers valid)

1.1.16 $\left(\sqrt{\sqrt{50} + x}\right)\left(\sqrt{\sqrt{50} - x}\right) = x$ (4)

Asked: 2019

1.1.17 $\sqrt{\sqrt{\frac{1}{x}} + 6} = \frac{1}{\sqrt{x}}$ (5)

Asked: 2025

1.1.18 $5^{x+2} - 5^{x+1} = 500$ (4)

Asked: 2016

1.1.19 $3^{2x} - 2 \cdot 3^{x+1} - 27 = 0$ (5)

Asked: 2024

1.1.20 $4 \cdot 2^{2x} - 33 \cdot 2^{x} + 8 = 0$ (4)

Asked: 2025

Given: $f(x) = x^2 + 2x - 15$

1.2.1 Solve for x if $f(x) = 0$. (2)

Asked: 2016

1.2.2 Solve for x if $f(x) < 0$. (2)

Asked: 2016

1.2.3 Determine the values of x for which $f'(x) \geq 0$. (2)

Asked: 2016

Given: $f(x) = x^2 + 10x + 25$

1.3.1 Solve for x if $f(x) > 0$. (3)

Asked: 2017

1.3.2 For which values of p will $f(x) = p$ have TWO unequal NEGATIVE roots? (4)

Asked: 2017

Solve for x and y simultaneously:

1.4.1 $x = 3y$ and $x^2 - 4xy = -27$ (4)

Asked: 2016

1.4.2 $2x - y = 5$ and $x^2 - xy - y^2 = -5$ (6)

Asked: 2017, 2020, 2021, 2024 (5 marks)

1.4.3 $y - 2x = 1$ and $x^2 - 3y = -8$ (6)

Asked: 2018

1.4.4 $2x + y = 7$ and $xy = 6$ (5)

Asked: 2019, 2022

1.4.5 $x + 3 = 3y$ and $\frac{3}{x} - \frac{1}{y} = \frac{1}{2}$ (5)

Asked: 2023

1.4.6 Calculate the values of x and y if: x is the sum of 3 and y; twice the product of x and y is 7 more than the square of x. (6)

Asked: 2025

Answer the following questions on exponents and number patterns.

1.5.1 Given: $3^{m+2} - 3^m = 2^{n+1} + 2^n$ where m and n are integers. Determine the value of $m + n$. (4)

Asked: 2023

1.5.2 Determine the values of x and y if: $8 \cdot 3^{y+1} = \sqrt{72^x}$ (4)

Asked: 2022

1.5.3 If n is the largest integer for which $n^{60} < 7^{90}$, determine the value of n. (3)

Asked: 2020

1.5.4 If $2^{4x} = 81$ and $7^{\sqrt{q}} = 81$, calculate, WITHOUT the use of a calculator, the value of: $\frac{[2^{x-2}]^4}{\sqrt{7^{\sqrt{q}}}}$ (4)

Asked: 2018

1.5.5 Show that $2 \cdot 3^n - 3^{n+1} + 3^{n+2}$ is even for all positive integer values of n. (3)

Asked: 2022

1.5.6 Consider the product $1 \times 2 \times 3 \times 4 \times \ldots \times 40$. Determine the largest value of k such that 5^k is a factor of this product. (4)

Asked: 2019

Consider the product $\left(1+\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{5}\right)\ldots\left(1+\frac{1}{n}\right)$, where each factor has the form $\left(1+\frac{1}{k}\right)$ for $k \geq 3$.

1.5.7 Determine ALL the values of n for which the product will be an integer. (3)

Asked: 2024

1.5.8 The roots of an equation are $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ where a, b and c are positive real numbers. The numbers a, b and c, in that order, form a geometric sequence. Prove that x is a non-real number. (4)

Asked: 2021

[149]

QUESTION 2: QUADRATIC PATTERNS [75 marks of variants]

Every year: a quadratic number pattern. Always: find the general term $T_n = an^2 + bn + c$ in some form (determine it, or show a given one). Usually: use T_n to find a term, or find which term has a given value (a quadratic equation, reject the negative root). Often: the first differences (their own general term, their sum, or which two terms differ by a given amount). Sometimes: the least or greatest term, a pattern with an unknown x in the terms, or a real-life context. 10 to 19 marks.

2.1 The first four terms of a quadratic number pattern are 2 ; x ; 12 ; $x + 18$.

2.1.1 Calculate the value of x. (4)

Asked: 2016

2.1.2 If $x = 5$, determine the position of the first term in the pattern for which the sum of the first n first differences will be greater than 500. (4)

Asked: 2016

2.2 The first four terms of a quadratic number pattern are: 192 ; 158 ; 128 ; 102 ; ...

2.2.1 Write down the next TWO terms of the pattern. (2)

Asked: 2018 (next term), 2019 (next two terms), 2021 (T4 from three terms)

2.2.2 Determine the constant second difference of the pattern. (2)

Asked: 2017

2.2.3 Determine the general term, $T_n = an^2 + bn + c$, of the pattern. (4)

Asked: 2017, 2018, 2019, 2020

2.2.4 Calculate the value of the 25th term of the pattern. (2)

Asked: 2018, 2020 (50th term), 2022 (60th term once b is known)

2.2.5 Which term of the pattern will be equal to 1830? (3)

Asked: 2017

2.2.6 Which term(s) of the pattern will have a value of 38? (4)

Asked: 2019

2.2.7 Which term in the pattern has the least value? (2)

Asked: 2019

2.3 Consider the quadratic number pattern: 1 ; 12 ; 31 ; 58 ; ...

2.3.1 Show that the sum of the first n first differences of this pattern can be given by $S_n = 4n^2 + 7n$. (3)

Asked: 2020

2.3.2 How many consecutive first differences were added to the first term of the pattern to obtain a term with a value of 6681? (4)

Asked: 2020

2.4 Consider the quadratic number pattern: -83 ; -68 ; -55 ; ... The general term of the pattern is $T_n = -n^2 + 18n - 100$.

2.4.1 Between which TWO terms of the pattern will there be a difference of -51? (4)

Asked: 2021

2.4.2 What value must be added to each term of the pattern so that the value of the maximum term of the new pattern will be 1? (3)

Asked: 2021

2.5 The general term of a quadratic number pattern is $T_n = 2n^2 + bn + 11$ and the first term of the first differences is 9.

2.5.1 Show that $b = 3$. (2)

Asked: 2022

2.5.2 Determine the general term of the sequence of first differences of the pattern. Write your answer in the form $T_p = mp + q$. (3)

Asked: 2022

2.5.3 Which TWO consecutive terms of the pattern have a first difference of 193? (3)

Asked: 2022

2.6 The following information is given about a quadratic number pattern: $T_1 = 4$, $T_2 - T_1 = 7$ and $T_3 - T_2 = 15$.

2.6.1 Show that $T_5 = 80$. (2)

Asked: 2023

2.6.2 Show that the general term of the pattern is $T_n = 4n^2 - 5n + 5$. (3)

Asked: 2023

2.6.3 Show that the pattern is increasing for all $n \in \mathbb{N}$. (3)

Asked: 2023

2.7 The sequence of the first differences of a quadratic number pattern is: 2 ; 6 ; 10 ; ...

2.7.1 If T_{80} of the pattern is $12\,499$, calculate the value of T_{79}. (3)

Asked: 2024

2.7.2 If it is further given that the third term of the pattern is 25, determine the general term, T_n, of the pattern. (5)

Asked: 2024

2.8 A flare is fired straight up from a ship. Its height above sea level (in metres) forms a quadratic pattern when it is measured every second. At the end of the 1st second the flare is 28 m high, at the end of the 2nd second it is 54 m high and at the end of the 3rd second it is 76 m high.

2.8.1 Calculate the height of the flare at the end of the 5th second. (2)

Asked: 2025

2.8.2 Show that the height of the flare at the end of n seconds is $T_n = -2n^2 + 32n - 2$. (3)

Asked: 2021, 2025

2.8.3 Calculate the maximum height that the flare reached. (3)

Asked: 2025 (also 2021 for a maximum term)

2.8.4 After how many seconds was the flare 76 m high for the second time? (2)

Asked: 2025

[75]

QUESTION 3: ARITHMETIC AND GEOMETRIC SEQUENCES AND SERIES [154 marks of variants]

Every year: arithmetic and/or geometric sequences and series. Arithmetic: find terms, the number of terms, a sum (often of a special set of terms), sometimes with an unknown k in the first three terms. Geometric: find r, a term, a finite sum, or a sum to infinity, often with a reason why the series converges ($-1 < r < 1$). Sigma notation is common: expand it, solve for an unknown limit, or use it to link two sums. Often a real-life or geometric context (strips on a sheet, steps, nested circles). 6 to 19 marks.

3.1 Given the finite arithmetic sequence: 62 ; 55 ; 48 ; ... ; -78 ; -85

3.1.1 Write down the fourth term (T_4) of the sequence. (1)
Asked: 2016

3.1.2 Calculate the number of terms in the sequence. (3)
Asked: 2016

3.1.3 Calculate the sum of all the negative numbers in the sequence. (3)
Asked: 2016

3.1.4 Consider the sequence: 62 ; 55 ; 48 ; ... ; -85 ; ... ; -2017. Determine the number of terms in this sequence that are exactly divisible by 5. (4)
Asked: 2016

3.2 -5 ; x ; y ; 22 ; ... is an arithmetic sequence.

3.2.1 Determine the values of x and y. (4)
Asked: 2020

3.3 The first three terms of an arithmetic sequence are $2k + 1$; $3k + 3$ and $5k + 2$.

3.3.1 Calculate the value of the 15th term of the sequence. (5)
Asked: 2017

3.3.2 Calculate the sum of the first 30 EVEN terms of the sequence. (4)
Asked: 2017

3.4 Given the arithmetic series: $9 + 13 + 17 + \ldots$

3.4.1 Determine the value of T_{85}. (3)
Asked: 2021 (T51), 2023 (T91)

3.4.2 Calculate S_{85}. (2)
Asked: 2021 (S51), 2023 (S91)

3.4.3 Calculate the value of n for which $T_n = 525$. (3)
Asked: 2018 (which term equals a given value), 2023

3.5 Given the arithmetic sequence 20 ; 17 ; 14 ; ... and the quadratic number pattern with general term $T_n = 2n^2 - 6n + 39$.

3.5.1 For which value of n will the sum of the first n terms of the arithmetic sequence be equal to the nth term of the quadratic pattern? (6)
Asked: 2018

3.6 The first term of an arithmetic series is 4. The common difference is 3 and the series contains 25 terms.

3.6.1 Calculate the sum of this series. (2)
Asked: 2024

3.6.2 The series is extended to 80 terms. The sum of these 80 terms is 9800. Using sigma notation, write down an equation for the sum of the terms that were added to the original series. (4)
Asked: 2024

3.7 Consider the sum: $\sum_{n=1}^{3000}(4n+1)$

3.7.1 Write down the expansion of this sum. Show only the first 3 terms and the last term of the expansion. (1)

Asked: 2021

3.7.2 Hence, or otherwise, calculate $\sum_{n=1}^{3000}(4n+1)+\sum_{n=1}^{2999}(-4n+3)$. ALL working details must be shown. (4)

Asked: 2021

3.8 Given: $\sum_{p=0}^{k}\left(\frac{1}{4}p+\frac{1}{8}\right)=10\frac{1}{8}$

3.8.1 Determine the value of k. (5)

Asked: 2022

3.9 Given: $\sum_{p=k}^{90}(3p+2)=11\,102$

3.9.1 Write down the difference between T_8 and T_{17}. (2)

Asked: 2025

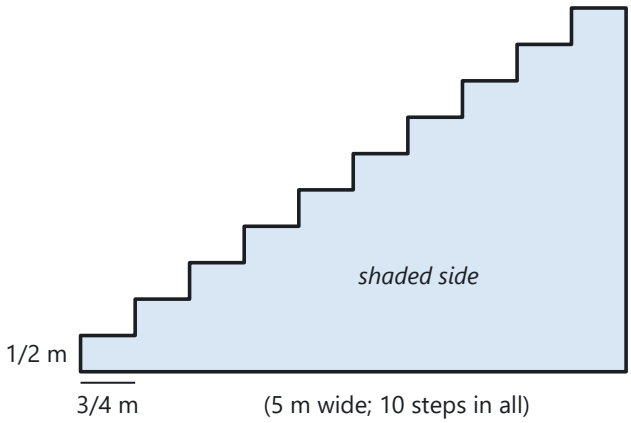
3.9.2 Calculate the value of k. (5)

Asked: 2025

3.10 Without using a calculator, determine the value of: $\sum_{y=4}^{12}\frac{1}{y-3}-\sum_{y=4}^{12}\frac{1}{y-2}$ (3)

Asked: 2019

3.11 A wooden stand at a sports ground has 10 steps. Each step is 5 m wide, has a rise of $\frac{1}{2}$ m and a tread (depth) of $\frac{3}{4}$ m, as shown in the side view below. The shaded side of the stand, on each side, must be covered with metal sheeting.



3.11.1 Calculate the area (in m^2) of metal sheeting needed to cover BOTH shaded sides. (6)

Asked: 2019

3.12 Given the geometric series: $x + 96 + 72 + \ldots$

3.12.1 Calculate the value of x. (2)

Asked: 2021

3.12.2 Show that the sum of the first n terms is $S_n = 512\left(1 - (0{,}75)^n\right)$. (2)

Asked: 2021

3.12.3 Hence, or otherwise, calculate the sum to infinity. (2)

Asked: 2021

3.13 A geometric series has a constant ratio of $\frac{1}{4}$ and a sum to infinity of 12.

3.13.1 Calculate the first term of the series. (2)

Asked: 2018

3.13.2 Calculate the 6th term of the series. (2)

Asked: 2018

3.13.3 Given: $\sum_{k=1}^{n} 9(4)^{1-k} = 11{,}8125$. Calculate the value of n. (4)

Asked: 2018

3.13.4 If $\sum_{k=1}^{20} 9(4)^{1-k} = p$, write down $\sum_{k=1}^{20} 144(4)^{-k}$ in terms of p. (3)

Asked: 2018

3.14 Given the geometric series: $4 + \frac{4}{3} + \frac{4}{9} + \ldots$

3.14.1 Calculate the sum of the first 6 terms of the series. (3)

Asked: 2019

3.14.2 Determine the largest value of n for which $T_n > \frac{4}{6561}$. (4)

Asked: 2019

3.15 The first term of a geometric series is 5 and the 6th term is 1215.

3.15.1 Calculate the value of the constant ratio, r. (2)

Asked: 2022

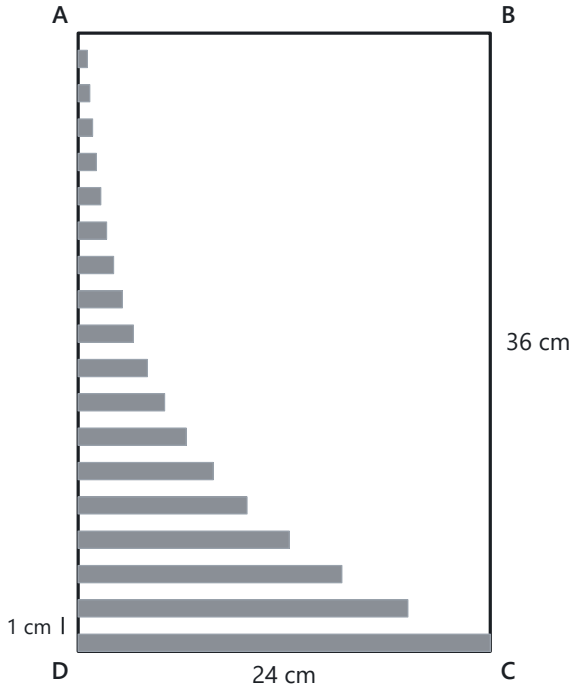
3.15.2 Determine the number of consecutive terms that must be added to the first 6 terms of the series in order to obtain a sum of $16\,400$. (4)

Asked: 2022

3.15.3 If the first term of another series is 1215 and its 6th term is 5, calculate the sum to infinity of the new series. (3)

Asked: 2022

3.16 Rectangles of width 1 cm are drawn from the edge of a sheet of paper that is 36 cm long such that there is a 1 cm gap between one rectangle and the next. The length of the first rectangle is 24 cm and the length of each successive rectangle is 80% of the length of the previous rectangle until there are rectangles drawn along the entire length of AD. Each rectangle is coloured grey.



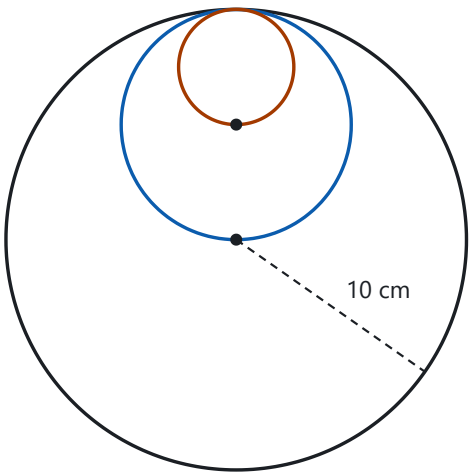
3.16.1 Calculate the length of the 10th rectangle. (3)

Asked: 2016

3.16.2 Calculate the percentage of the paper that is coloured grey. (4)

Asked: 2016

3.17 A circle with radius 10 cm is drawn. A second, smaller circle is drawn through the centre of the first circle and also touches the first circle internally, as shown in the diagram. A third, smaller circle is drawn through the centre of the second circle and touches the second circle internally. The process of drawing circles continues and forms a geometric pattern. The dots show the centres of the first and second circles.



3.17.1 Write down the radius of the third circle. (2)

Asked: 2024

3.17.2 Calculate the sum of the areas of the first 8 circles. (4)

Asked: 2024

3.17.3 Which circle has a diameter of $\frac{5}{64}$ cm? (4)

Asked: 2024

3.18 A convergent geometric series consisting of only positive terms has first term a, constant ratio r and n^{th} term, T_n, such that $\sum_{n=3}^{\infty} T_n = 4$.

3.18.1 If $T_1 + T_2 = 5$, write down an expression for a in terms of r. (2)

Asked: 2017

3.18.2 Calculate the values of a and r. (6)

Asked: 2017

3.19 Given: $\sum_{k=1}^{\infty} 5(2)^{3-k}$

3.19.1 Prove that this is a convergent geometric series. Show ALL your calculations. (3)

Asked: 2020

3.19.2 If $\sum_{k=p}^{\infty} 5(2)^{3-k} = \frac{5}{8}$, determine the value of p. (5)

Asked: 2020

3.20 Given the geometric series: $5 + 15 + 45 + \ldots$ to n terms.

3.20.1 Write down the general term of this series. (1)

Asked: 2023

3.20.2 Calculate the value of k such that: $\sum_{p=1}^{k} \frac{5}{3}(3)^p = 49\,205$ (4)

Asked: 2023

3.20.3 A geometric sequence and an arithmetic sequence have the same first term. The common ratio of the geometric sequence is $\frac{1}{4}$. The common difference of the arithmetic sequence is 2. The sum of 20 terms of the arithmetic sequence is 436 more than the sum to infinity of the geometric sequence. Calculate the value of the first term. (5)

Asked: 2023

3.21 Given the infinite geometric series: $(t + 32) + (t - 4) + (t + 8) + \ldots$

3.21.1 Show that $t = -5$. (3)

Asked: 2025

3.21.2 Calculate the value of T_{25}. Write your answer in the form $T_n = b^x$. (3)

Asked: 2025

3.21.3 Calculate the sum of the infinite series. (2)

Asked: 2025

[154]

QUESTION 4: FUNCTIONS: HYPERBOLA

[47 marks of variants]

Every year: a hyperbola $f(x) = \frac{a}{x+p} + q$, either given as an equation or drawn with its asymptotes. Always: the equations of the asymptotes (or the values of p and q) and a way of finding a. Usually: the domain or range, the intercepts and a sketch. Often: an axis of symmetry $y = \pm x + c$ through the point where the asymptotes meet, a sign question such as $f(x) \leq 0$ or $x.f'(x) \geq 0$, or a transformation (a translation such as $h(x+3)$, or a reflection). Sometimes the hyperbola is given as $(x-p)(y+t) = k$ or $y = \frac{d}{x}$. 8–10 marks when asked alone, up to 20 marks inside a longer functions question.

Given: $f(x) = \frac{-2}{x-3} + 1$

4.1 Write down the equations of the asymptotes of f. (2)

Asked: 2018, 2020, 2021

4.2 Write down the domain of f. (1)

Asked: 2018, 2021, 2025

4.3 Write down the range of f. (1)

Asked: 2017, 2023

4.4 Determine the coordinates of the x-intercept of f. (2)

Asked: 2021, 2022, 2025

4.5 Determine the coordinates of the y-intercept of f. (2)

Asked: 2021

4.6 Draw the graph of f. Clearly show ALL the asymptotes and the intercepts with the axes. (3)

Asked: 2018 (intercepts only), 2020, 2021

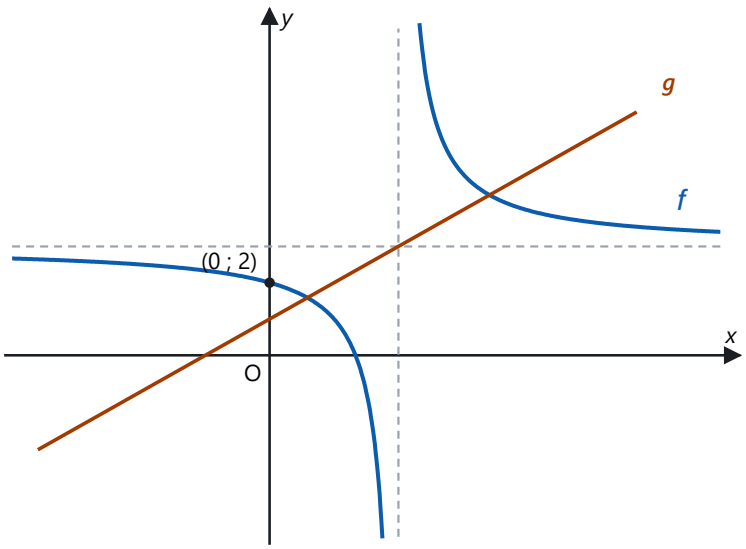
4.7 For which values of x will $x.f'(x) \geq 0$? (2)

Asked: 2018

4.8 Determine the values of x for which $\frac{-2}{x-3} \geq -1$. (3)

Asked: 2022

The graph of $f(x) = \frac{a}{x+p} + q$ is sketched below. The domain of f is $(-\infty; 2) \cup (2; \infty)$. The graph of f cuts the y-axis at $(0; 2)$. The straight line $g(x) = x + 1$ is an axis of symmetry of f.



4.9 Write down the value of p. (1)

Asked: 2019, 2024

4.10 Determine the equation of the horizontal asymptote of f. (2)

Asked: 2024

4.11 Calculate the value of a. (2)

Asked: 2019, 2023 (d), 2024

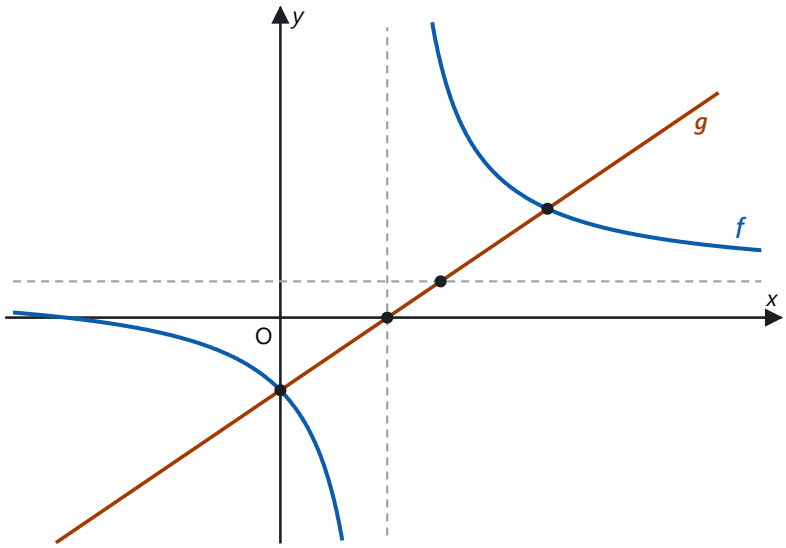
4.12 For which values of x is $f(x) \leq 0$? (3)

Asked: 2024

4.13 The graph of f is transformed to the graph of h. The domain and the range of h are the same as those of f, and $h'(x)$ is positive on its domain. Describe a possible transformation of f that gives h. (2)

Asked: 2024

The graphs of $g(x) = x + c$ and $f(x) = \dfrac{a}{x+p} + q$ are drawn below. The graph of g and the vertical asymptote of f intersect on the x-axis.



4.14 Write down the coordinates of the x-intercept of g in terms of p. (1)

Asked: 2025

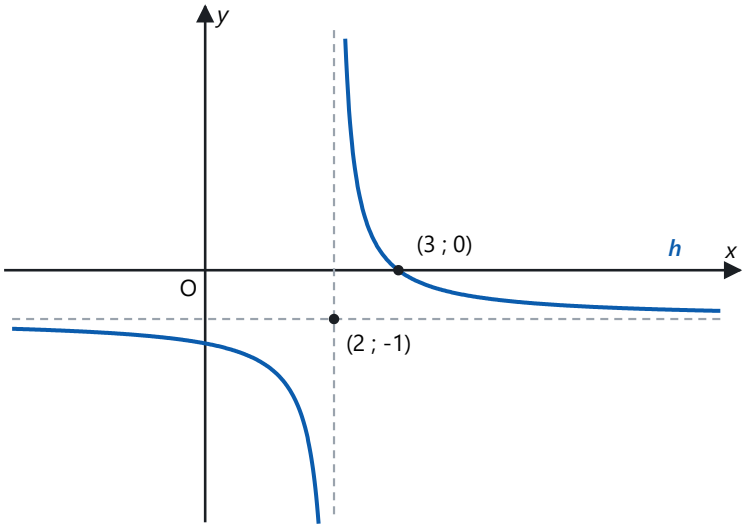
4.15 The graph of g intersects the horizontal asymptote of f at $x = 3$ and the graph of f at $x = 5$. The graphs of f and g also intersect on the y-axis. Determine the equation of f. (5)

Asked: 2025

4.16 Describe the transformation that g must undergo to become an axis of symmetry of f that cuts f at two points. (2)

Asked: 2025

Sketched below is the graph of $h(x) = \frac{1}{x+p} + q$. The asymptotes of h intersect at $(2; -1)$. The graph of h cuts the x-axis at $(3; 0)$.



4.17 Write down the values of p and q. (2)

Asked: 2022

4.18 Determine the equation of the axis of symmetry of h that has a NEGATIVE gradient. (2)

Asked: 2020, 2022 (y = x + t), 2017 (first step)

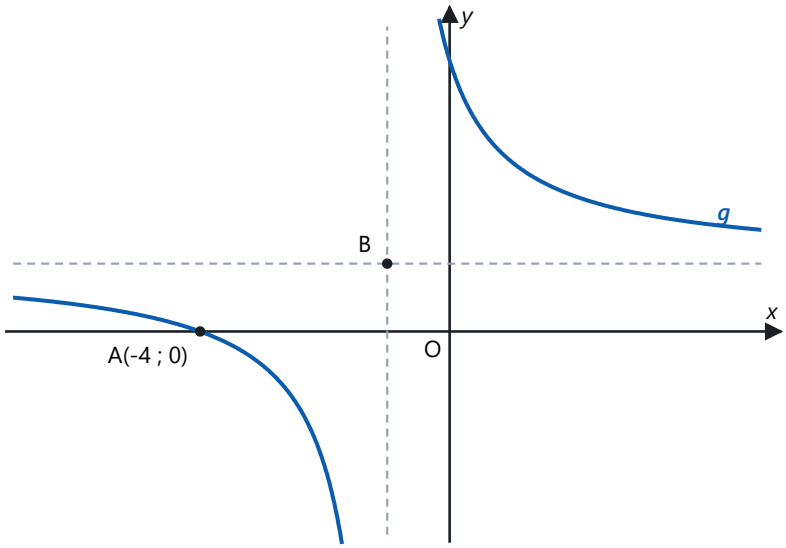
4.19 The graph of k is obtained when h is translated 4 units to the left, i.e. $k(x) = h(x+4)$. (1)
Write down the equation of the vertical asymptote of k.

Asked: 2016

4.20 Determine the x-coordinate of the x-intercept of g, where $g(x) = h(x+2)$. (2)

Asked: 2022

The graph of g has the equation $(x-p)(y+t) = 6$ and is sketched below. The asymptotes of g intersect at $B(-1; 2)$ and $A(-4; 0)$ is the x-intercept of g.



4.21 For which values of x will $g(x) \geq 0$? (2)

Asked: 2016

4.22 Determine the values of p and t. (4)

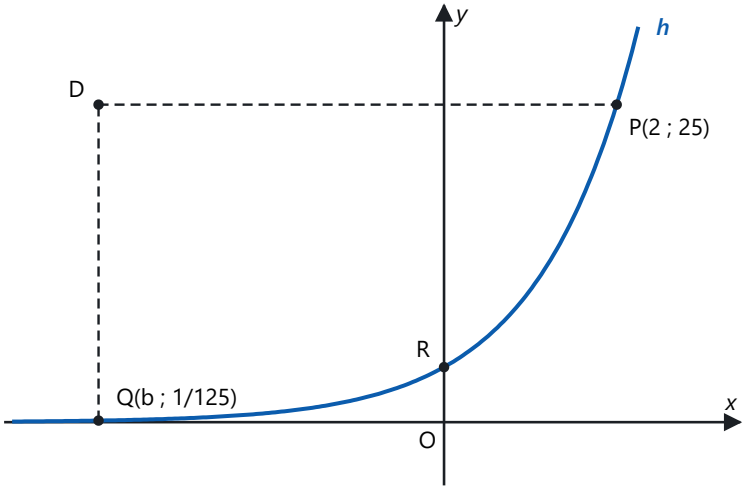
Asked: 2016

[47]

QUESTION 5: FUNCTIONS: EXPONENTIAL, LOGARITHMIC AND INVERSE [100 marks of variants]

Every year: an exponential graph $y = a^x$ (sometimes shifted, reflected or with a restricted domain) or a logarithmic graph $y = \log_a x$. Always: find a (or k) from a given point, plus intercepts or an asymptote. Usually: the inverse function (swap x and y), its domain or range, and a sketch. Often: an inequality, a transformation, an equation solved from the graph, or a hyperbola or straight line on the same set of axes. Occasionally: a hard equation such as $2^x - 2^{-x} = c$. 8–16 marks.

Sketched below is the graph of $h(x) = a^x$, $a > 0$. R is the y-intercept of h. The points $P(2; 25)$ and $Q\left(b; \frac{1}{125}\right)$ lie on h.



5.1 Determine the coordinates of R. (1)

Asked: 2016, 2020

5.2 Calculate the value of a. (2)

Asked: 2016, 2019, 2024

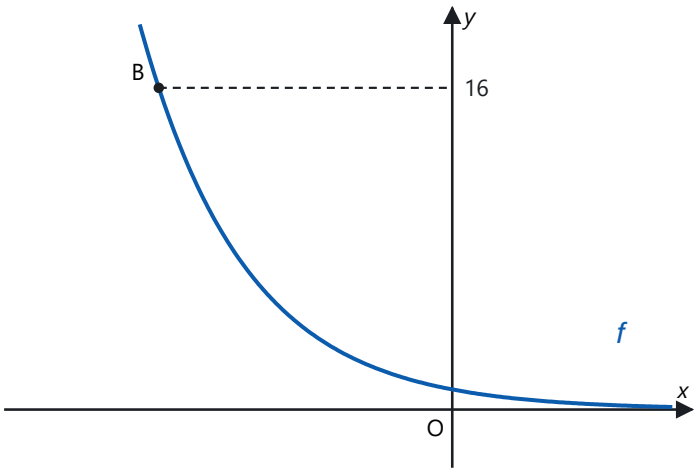
5.3 D is a point such that $DQ \parallel y$-axis and $DP \parallel x$-axis. Calculate the length of DP. (4)

Asked: 2016

5.4 Determine the values of k for which the equation $h(x + 1) + k = 0$ will have a root that is less than -4. (3)

Asked: 2016

The graph of $f(x) = 2^{-x}$ is sketched below. B is the point of intersection of f and the line $y = 16$.



5.5 Determine the coordinates of B. (3)

Asked: 2020

5.6 Write down the domain of f^{-1}, the inverse of f. (2)

Asked: 2020

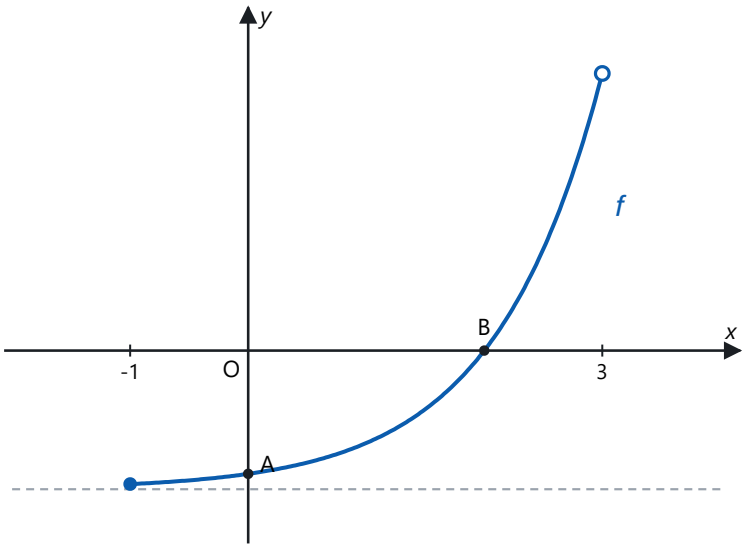
5.7 Describe the translation from f to $g(x) = \frac{32}{2^x}$. (3)

Asked: 2020

5.8 Determine the values of x for which $g(x) < 1$. (3)

Asked: 2020

Sketched below is the graph of $f(x) = 3^x - 9$ for $x \in [-1; 3)$. A and B are respectively the y- and the x-intercepts of f.



5.9 Write down the equation of the asymptote of f. (1)

Asked: 2016 (y = 0), 2023

5.10 Determine the coordinates of B. (2)

Asked: 2023

5.11 Determine the equation of k, the straight line through A and B, in the form $k(x) = \ldots$ (3)

Asked: 2023

5.12 Calculate the vertical distance between k and f at $x = 1$. (3)

Asked: 2023

5.13 The graph of g is obtained by moving f up by 9 units. Write down the equation of g, with its domain. (1)

Asked: 2023

5.14 Write down the domain of g^{-1}, the inverse of g. (2)

Asked: 2023

5.15 Write down the equation of g^{-1} in the form $y = \ldots$ (2)

Asked: 2023

Given: $f(x) = a^x - 4$ for $a > 0$. The point $B\left(2; -\frac{15}{4}\right)$ lies on f.

5.16 Calculate the value of a. (2)

Asked: 2024

5.17 Write down the range of f. (1)

Asked: 2024

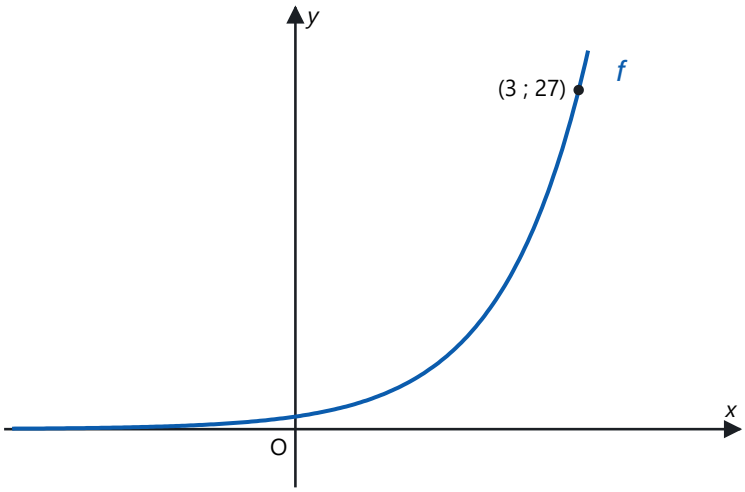
5.18 Sketch the graph of f. Clearly show the intercepts with the axes and the asymptote. (3)

Asked: 2024

5.19 C is the point on f where $y = 4$. Determine the coordinates of C', the image of C when C is reflected about the line $y = x$. (3)

Asked: 2024

Sketched below is the graph of $f(x) = 3^x$. The point $(3; 27)$ lies on f.



5.20 The graph g is obtained by reflecting f about the line $y = x$. Determine the equation of g in the form $y = \ldots$ (2)

Asked: 2019

5.21 Sketch the graph of g. Indicate the coordinates of TWO points on g. (4)

Asked: 2019

5.22 Use your graph to determine the values of x for which $f(x) \cdot g(x) > 0$. (2)

Asked: 2019

5.23 Use your graph to determine the values of x for which $g(x) \leq -1$. (2)

Asked: 2019

5.24 If $h(x) = f(-x)$, calculate the value of x for which $f(x) - h(x) = \frac{8}{3}$. (4)

Asked: 2019

Given: $f(x) = -x + 4$ and $g(x) = \log_3 x$

5.25 On the same set of axes, sketch the graphs of f and g. Clearly show ALL the intercepts with the axes. (4)

Asked: 2016

5.26 Write down the equation of g^{-1}, the inverse of g, in the form $y = \ldots$ (2)

Asked: 2016, 2017, 2021

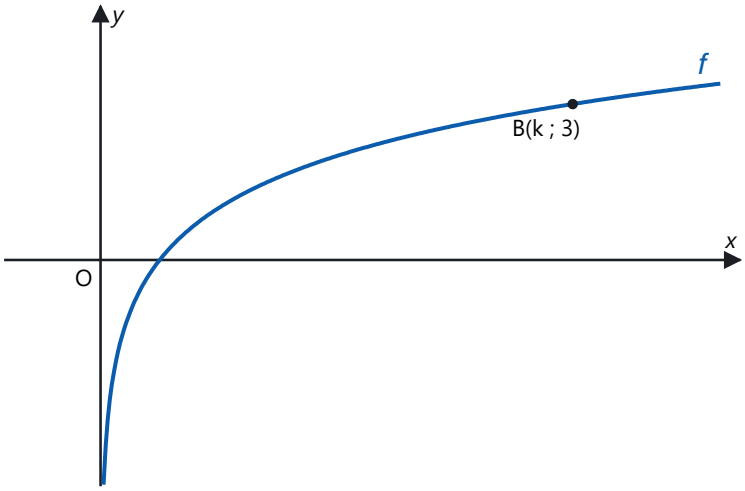
5.27 Explain how you will use your graphs in QUESTION 5.25 and/or the equation in QUESTION 5.26 to solve the equation $\log_3(4 - x) = x$. (3)

Asked: 2016

5.28 Write down the solution to $\log_3(4 - x) = x$. (1)

Asked: 2016

The graph of $f(x) = \log_2 x$ is drawn below. $B(k; 3)$ is a point on f.



5.29 Calculate the value of k. (2)

Asked: 2021

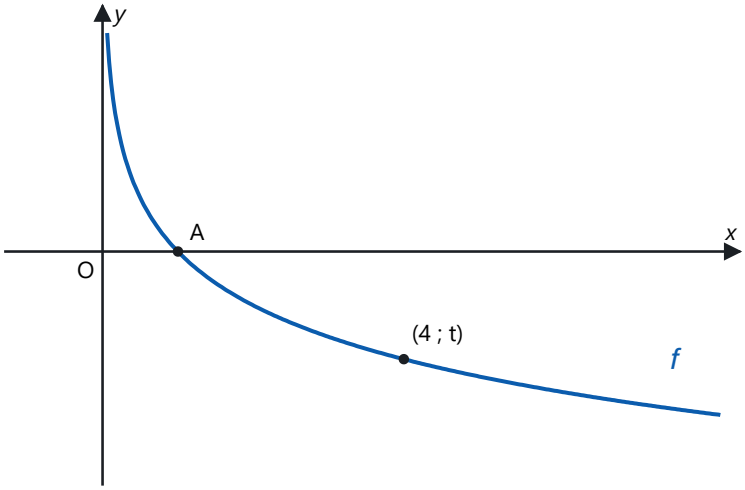
5.30 Determine the values of x for which $-2 \leq f(x) \leq 3$. (2)

Asked: 2021

5.31 For which values of x will $x \cdot f^{-1}(x) < 0$, where f^{-1} is the inverse of f? (2)

Asked: 2021

The graph of $f(x) = \log_{1/2} x$ is drawn below. Point A is the x-intercept of f and $(4; t)$ lies on f.



5.32 Calculate the value of t. (1)

Asked: 2025

5.33 Write down the coordinates of A. (1)

Asked: 2025

5.34 Determine the equation of f^{-1}, the inverse of f, in the form $y = \ldots$ (2)

Asked: 2025

5.35 Write down the equation of the asymptote of f^{-1}. (1)

Asked: 2025

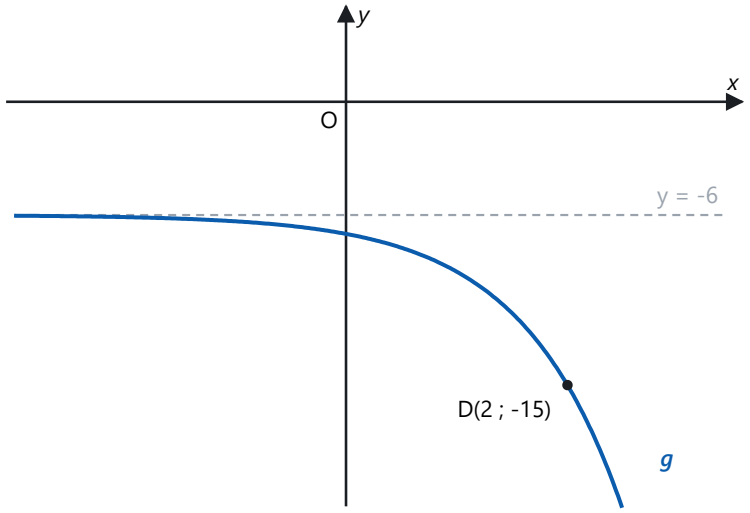
5.36 Draw the graph of f^{-1}. Clearly show the intercept with the axes, the coordinates of ONE other point and the asymptote. (3)

Asked: 2025

5.37 The graph of q is obtained when f^{-1} is translated 2 units to the right. Determine the y-values of q where $x > 3$. (2)

Asked: 2025

Sketched below is the graph of $g(x) = a \cdot 3^x + q$. The horizontal asymptote of g is the line $y = -6$ and the point $D(2; -15)$ lies on g.



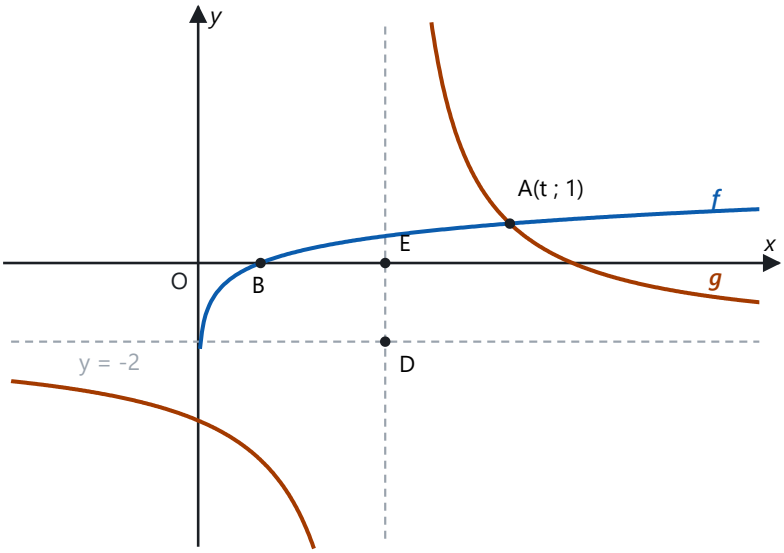
5.38 Determine the values of a and q. (3)

Asked: 2022

5.39 Write down the range of g. (1)

Asked: 2022

The diagram below shows the graphs of $g(x) = \dfrac{6}{x+p} + q$ and $f(x) = \log_5 x$. The line $y = -2$ is the horizontal asymptote of g. $B(1; 0)$ is the x-intercept of f. $A(t; 1)$ is a point of intersection of f and g. The vertical asymptote of g intersects the x-axis at E and the horizontal asymptote at D. $OE = 3 \cdot OB$.



5.40 Write down the range of g. (2)

Asked: 2017

5.41 Determine the equation of g. (2)

Asked: 2017

5.42 Calculate the value of t. (3)

Asked: 2017

5.43 For which values of x will $f^{-1}(x) < 25$? (2)

Asked: 2017

5.44 Determine the point of intersection of the graph of f and the axis of symmetry of g that has a NEGATIVE gradient. (3)

Asked: 2017

[100]

QUESTION 6: FUNCTIONS: PARABOLA, STRAIGHT LINE AND MIXED GRAPHS [167 marks of variants]

Every year: a parabola drawn together with a straight line, a hyperbola or an exponential graph. Always: intercepts, the turning point or the range of the parabola, and at least one tangent, distance or inequality question. Usually: the equation of the line or the value of a constant, a vertical distance, and reading values of x off the graphs. Sometimes: the inverse of a function, an area, a transformation, or the values of k for which two graphs do or do not meet. 8–23 marks.

6.1 Given: $f(x) = -ax^2 + bx + 12$, where a and b are constants.

6.1.1 The gradient of the tangent to the graph of f at the point $(-1\,;\,7)$ is 6. Show that $a = 1$ and $b = 4$. (5)

Asked: 2017

6.1.2 Calculate the x-intercepts of f. (3)

Asked: 2017, 2018

6.1.3 Calculate the coordinates of the turning point of f. (3)

Asked: 2017, 2019, 2021, 2024

6.1.4 Sketch the graph of f. Clearly show ALL intercepts with the axes and the turning point. (4)

Asked: 2017

6.1.5 Use your graph to determine the values of x for which $f(x) > 12$. (3)

Asked: 2017

6.1.6 Sketch the graph of $g(x) = -2x + 4$ on the same set of axes as f. Clearly show ALL intercepts with the axes. (2)

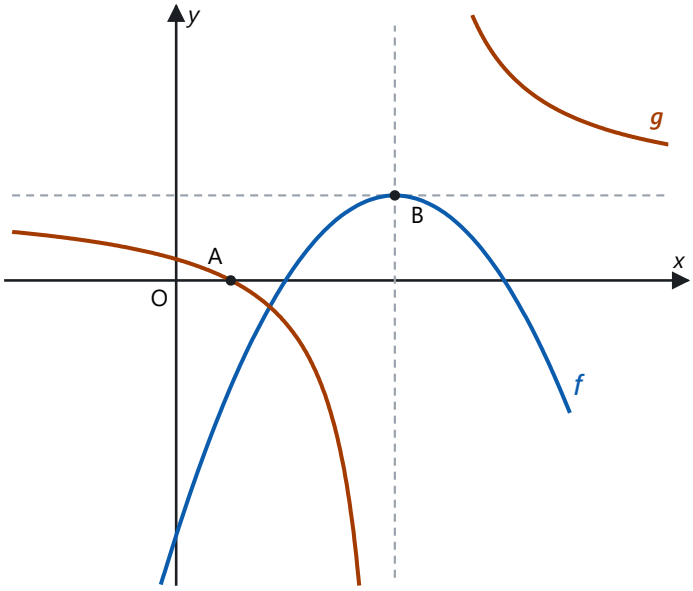
Asked: 2017

6.1.7 Write down the values of x for which $f(x) \cdot g(x) \leq 0$. (3)

Asked: 2017

6.2 Sketched below is the parabola f with equation $f(x) = -x^2 + 8x - 12$ and a hyperbola g with equation $(x - p)(y + t) = 12$.

- B, the turning point of f, lies at the point of intersection of the asymptotes of g.
- A$(1\,;\,0)$ is the x-intercept of g.



6.2.1 Show that the coordinates of B are $(4\,;\,4)$. (2)

Asked: 2016

6.2.2 Write down the range of f. (1)

Asked: 2016, 2019, 2020, 2021, 2025 (written down)

6.2.3 For which values of x will $g(x) \geq 0$? (2)

Asked: 2016

6.2.4 Determine the equation of the vertical asymptote of the graph of h if $h(x) = g(x+3)$. (1)

Asked: 2016

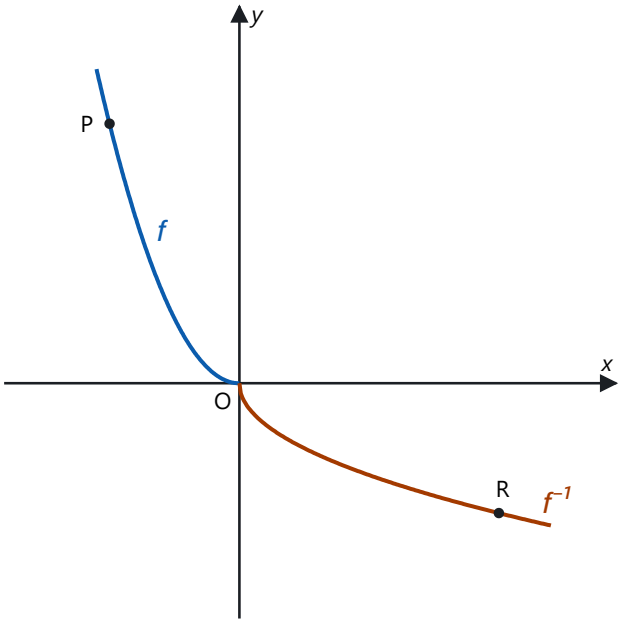
6.2.5 Determine the values of p and t. (4)

Asked: 2016

6.2.6 Write down the values of x for which $f(x) \cdot g'(x) \geq 0$. (4)

Asked: 2016

6.3 In the diagram below, the graph of $f(x) = ax^2$ is drawn for $x \leq 0$. The graph of f^{-1}, the inverse of f, is also drawn. P$(-4\,;\,8)$ is a point on f and R is a point on f^{-1}.



6.3.1 Is f^{-1} a function? Motivate your answer. (2)

Asked: 2018

6.3.2 R is the reflection of P in the line $y = x$. Write down the coordinates of R. (1)

Asked: 2018

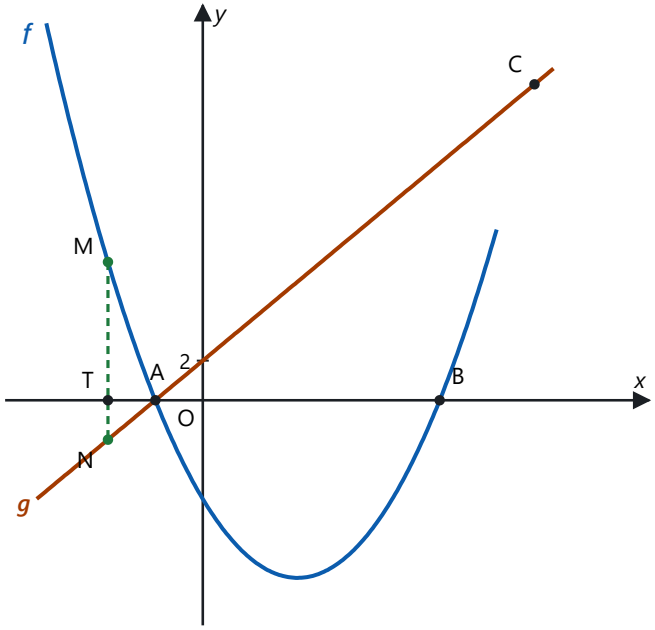
6.3.3 Calculate the value of a. (2)

Asked: 2018

6.3.4 Write down the equation of f^{-1} in the form $y = \ldots$ (3)

Asked: 2018

6.4 In the diagram below, A$(-1\,;\,0)$ and B$(5\,;\,0)$ are the x-intercepts of the graph of $f(x) = x^2 - 4x - 5$. A straight line, g, through A cuts f at C$(7\,;\,16)$ and the y-axis at $(0\,;\,2)$. M is a point on f and N is a point on g such that MN is parallel to the y-axis. MN cuts the x-axis at T.



6.4.1 Show that $g(x) = 2x + 2$. (2)

Asked: 2018

6.4.2 Determine the range of f. (3)

Asked: 2018

6.4.3 If MN = 9, calculate the length of OT if T lies on the negative x-axis. Show ALL your working. (4)

Asked: 2018

6.4.4 Hence write down the coordinates of N. (2)

Asked: 2018

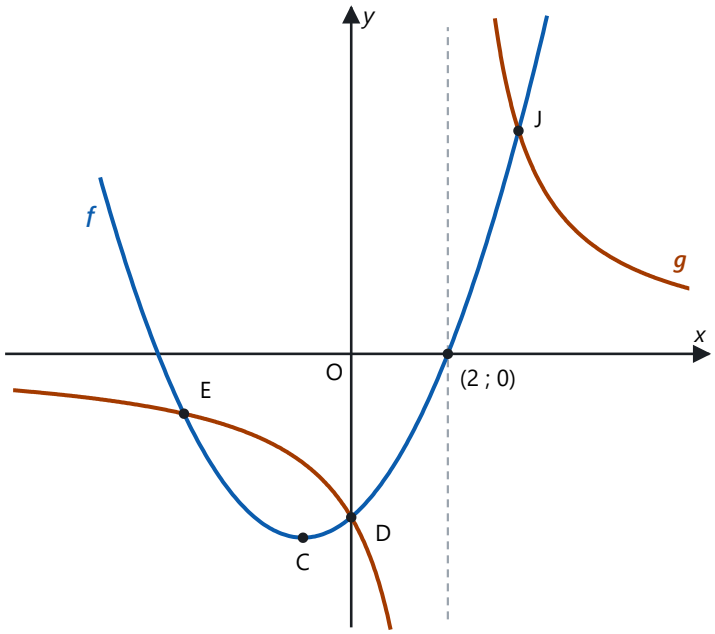
6.4.5 Determine the equation of the tangent to f that is parallel to g. (5)

Asked: 2018

6.4.6 For which value(s) of k will $f(x)$ and $h(x) = 2x + k$ NOT intersect? (1)

Asked: 2018

6.5 Below are the graphs of $f(x) = x^2 + bx - 8$ and $g(x) = \dfrac{a}{x+p}$. f has a turning point at C $(-1\,;\,-9)$ and passes through the x-axis at $(2\,;\,0)$. D is the y-intercept of both f and g. The graphs of f and g also intersect each other at E and J. The vertical asymptote of g passes through the x-intercept of f at $(2\,;\,0)$.



6.5.1 Write down the value of p. (1)

Asked: 2019

6.5.2 Show that $a = 16$ and $b = 2$. (3)

Asked: 2019

6.5.3 Determine the equation of the line through C that makes an angle of 45° with the positive x-axis. Write your answer in the form $y = \ldots$ (3)

Asked: 2019

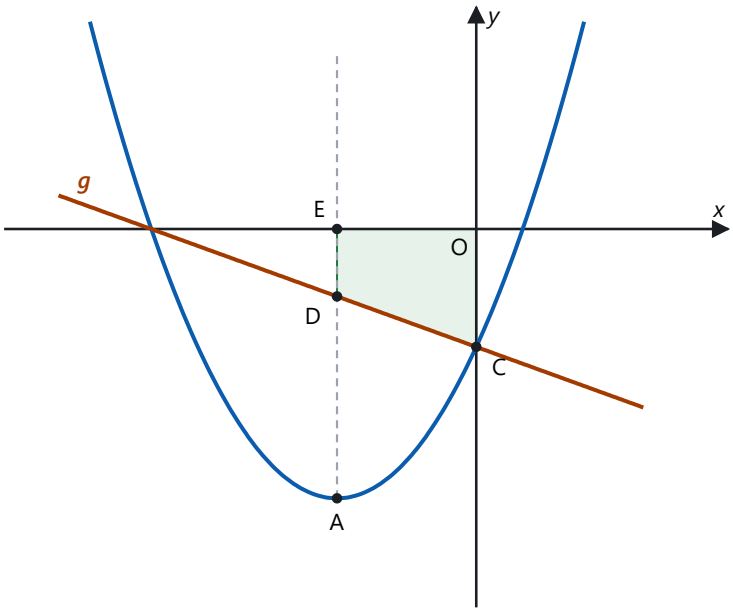
6.5.4 Is the straight line from QUESTION 6.5.3 a tangent to f? Explain your answer. (2)

Asked: 2019

6.5.5 The function $h(x) = f(m - x) + q$ has only one x-intercept, at $x = 0$. Determine the values of m and q. (4)

Asked: 2019

6.6 The graphs of $f(x) = \frac{1}{2}(x+3)^2 - 8$ and $g(x) = -\frac{1}{2}x - \frac{7}{2}$ are sketched below. A is the turning point of f. The axis of symmetry of f intersects the x-axis at E and the line g at D$(m\,;\,n)$. C is the y-intercept of both f and g.



6.6.1 Write down the coordinates of A, the turning point of f. (2)

Asked: 2020, 2023

6.6.2 Calculate the values of m and n. (3)

Asked: 2020

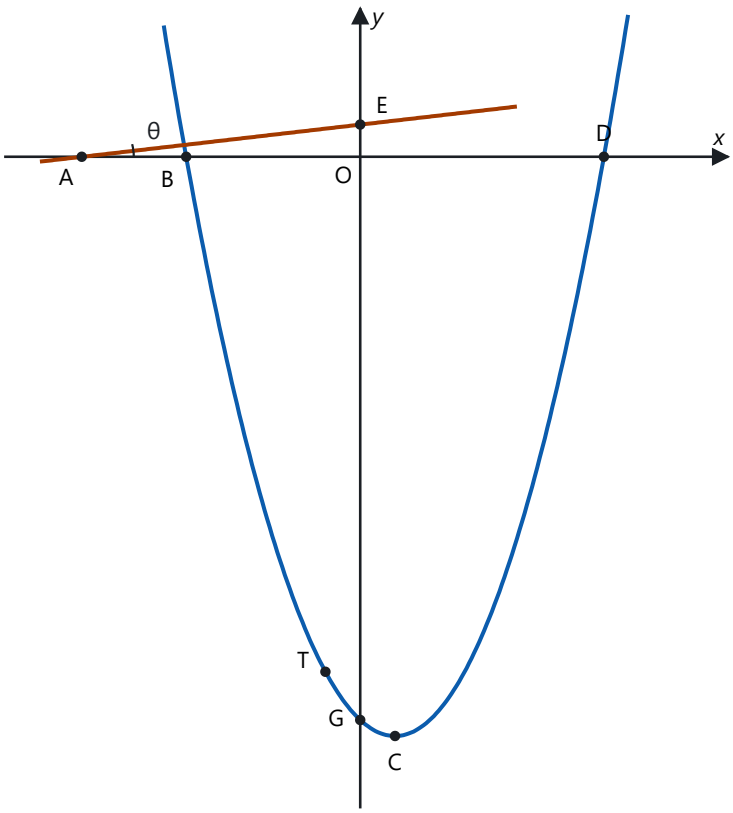
6.6.3 Calculate the area of OCDE. (3)

Asked: 2020

6.6.4 If $h(x) = g^{-1}(x) + k$ is a tangent to f, determine the coordinates of the point of contact between h and f. (5)

Asked: 2020

6.7 The graph of $f(x) = (x + 5)(x - 7)$ is drawn below. The parabola cuts the x-axis at B and D and the y-axis at G. C is the turning point of f. Line AE has an angle of inclination of θ and cuts the x-axis and y-axis at A and E respectively. T is a point on f between B and G.



6.7.1 Write down the coordinates of B and D. (2)

Asked: 2021

6.7.2 Given that $\theta = 14{,}04°$ and that the tangent to f at T is perpendicular to AE, calculate the gradient of AE, correct to TWO decimal places. (1)

Asked: 2021

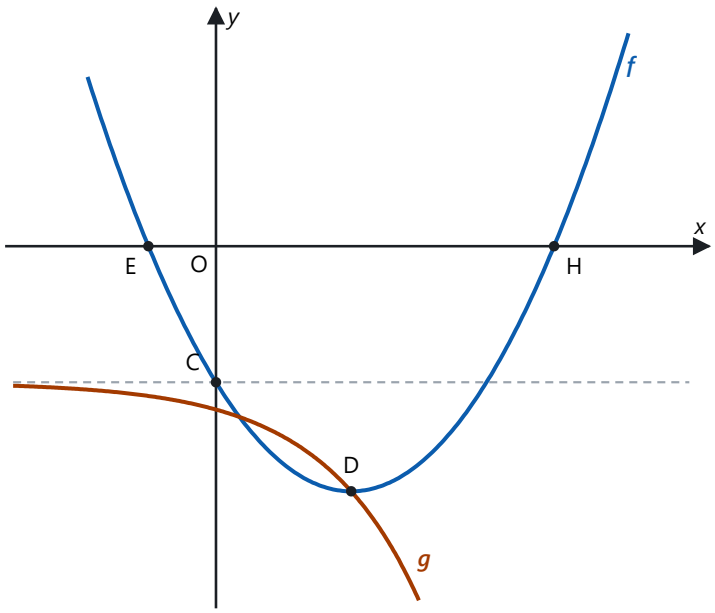
6.7.3 Calculate the coordinates of T. (5)

Asked: 2021

6.7.4 A straight line, g, parallel to AE, cuts f at K$(-2\,;\,-27)$ and at R. Calculate the x-coordinate of R. (6)

Asked: 2021

6.8 The graphs of $f(x) = 2x^2 - 8x - 10$ and $g(x) = a \cdot 2^x + q$ are sketched below. E and H are the x-intercepts of f. C is the y-intercept of f and lies on the asymptote of g. The graph of g passes through D, the turning point of f.



6.8.1 Write down the y-coordinate of C. (1)

Asked: 2022

6.8.2 Determine the coordinates of D. (2)

Asked: 2022

6.8.3 Determine the values of a and q. (3)

Asked: 2022

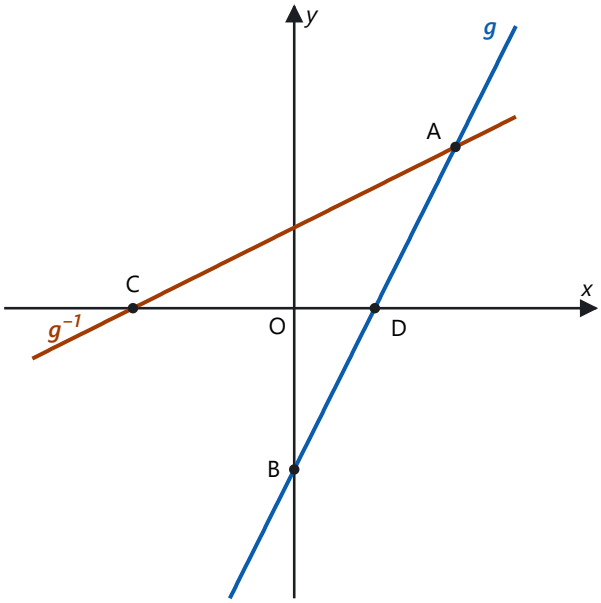
6.8.4 Write down the range of g. (1)

Asked: 2022

6.8.5 Determine the values of k for which $f(x) - k$ will always be positive. (2)

Asked: 2022

6.9 The graphs of $g(x) = 2x - 4$ and g^{-1}, the inverse of g, are shown in the diagram below. D and B are the x- and y-intercepts of g respectively. C is the x-intercept of g^{-1}. The graphs of g and g^{-1} intersect at A.



6.9.1 Write down the y-coordinate of B. (1)

Asked: 2022

6.9.2 Determine the equation of g^{-1} in the form $g^{-1}(x) = mx + n$. (2)

Asked: 2020, 2022

6.9.3 Determine the coordinates of A. (3)

Asked: 2022

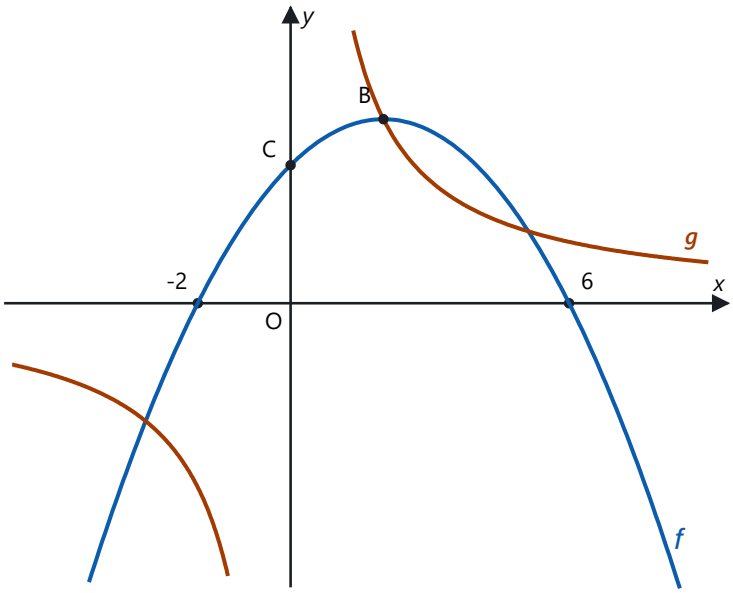
6.9.4 Calculate the length of AB. (2)

Asked: 2022

6.9.5 Calculate the area of ΔABC. (5)

Asked: 2022

6.10 The graphs of $f(x) = -\frac{1}{2}(x-2)^2 + 8$ and $g(x) = \frac{d}{x}$ are drawn below. A point of intersection of f and g is B, the turning point of f. The graph of f has x-intercepts at $(-2\,;\,0)$ and $(6\,;\,0)$ and a y-intercept at C.



6.10.1 Calculate the coordinates of C. (2)

Asked: 2023

6.10.2 Calculate the value of d. (1)

Asked: 2023

6.10.3 For which values of x will $f(x) \cdot g(x) \leq 0$? (3)

Asked: 2023 (with a hyperbola)

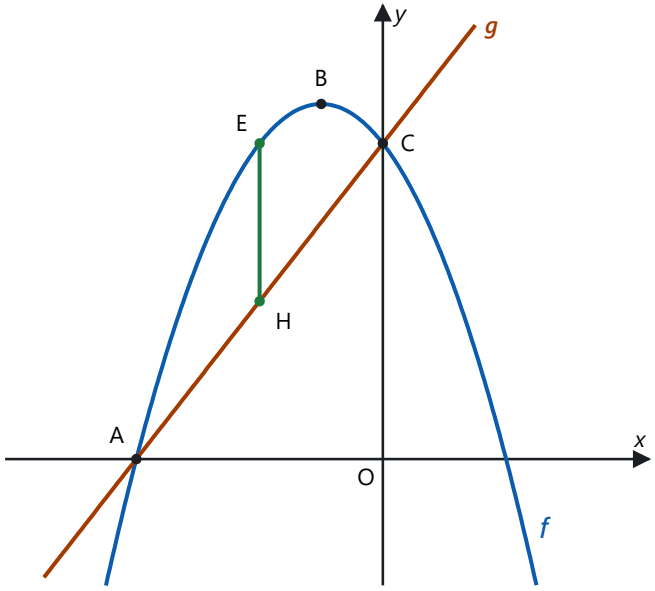
6.10.4 Calculate the values of k so that $h(x) = -x + k$ will NOT intersect the graph of g. (5)

Asked: 2023

6.10.5 $h(x) = -x + k$ is a tangent to g at R, a point in the first quadrant. Calculate the value of t such that $y = f(x) + t$ passes through R. (4)

Asked: 2023

6.11 In the diagram below, the graphs of $f(x) = -x^2 - 2x + 8$ and g, a straight line, are drawn. B is the turning point of f and A is the smaller x-intercept of f. C$(0\,;\,8)$ is a point of intersection of f and g. EH is drawn parallel to the y-axis, with E a point on f and H a point on g.



6.11.1 Show that the equation of the line through A and C is $g(x) = 2x + 8$. (3)

Asked: 2024

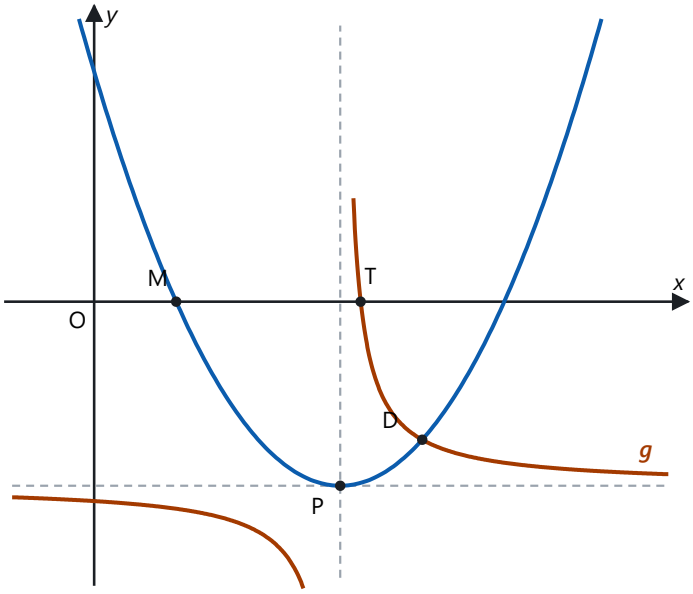
6.11.2 Calculate the maximum length of EH for $f > g$. (4)

Asked: 2024

6.11.3 Given: $k(x) = f(x + m) = -x^2 - 2mx - m^2 - 2x - 2m + 8$. Determine the value of m for which g is a tangent to k. (5)

Asked: 2024

6.12 The graphs of $f(x) = ax^2 + bx + c$ and $g(x) = \dfrac{2}{x - 3} - 8$ are drawn below. P is the turning point of f and the point of intersection of the asymptotes of g. The graphs intersect at D$(4\,;\,-6)$. M is the smaller x-intercept of f and T is the x-intercept of g.



6.12.1 Determine the values of x for which $g(x) > f(x)$. (2)

Asked: 2025

6.12.2 Determine the values of x for which $f(x) < -6$. (2)

Asked: 2025

6.12.3 Show that the equation of the parabola is $f(x) = 2x^2 - 12x + 10$. (3)

Asked: 2025

6.12.4 Calculate the length of MT. (6)

Asked: 2025

6.12.5 Determine the equation of the tangent to f at D. (3)

Asked: 2025

[167]

QUESTION 7: FINANCE, GROWTH AND DECAY [133 marks of variants]

Every year: compound interest and a loan or savings annuity with the formulae $A = P(1+i)^n$, $F = \frac{x[(1+i)^n - 1]}{i}$ and $P = \frac{x[1-(1+i)^{-n}]}{i}$. Every year: at least one "find n" (logs) or "find r" (roots) item. Usually: depreciation (reducing balance or straight line), an outstanding balance after some payments, and the interest paid over part of a loan. Often a twist: missed or deferred first payments, a larger instalment so the loan ends early, a final smaller payment, a lump sum paid into the loan, or the effective rate. 14–16 marks per year.

7.1.1 A school tour costs R36 500 now. The tour company predicts that the cost will increase by 6,4% per annum. What will the tour cost in 4 years' time? (2)

Asked: 2025

7.1.2 Lerato's grandparents deposited R8 000 into a savings account on the day she was born. The account pays interest at 5,6% p.a., compounded quarterly. Calculate the accumulated amount in this account on Lerato's 18th birthday. (3)

Asked: 2024

7.1.3 A farmer bought a combine harvester for R640 000. Its value depreciates at 8,6% p.a. on the reducing-balance method. Calculate the book value of the harvester after 6 years. (3)

Asked: 2021

7.1.4 A car was bought for R320 000. Its value depreciated at 18% p.a. on the reducing-balance method. After how many years was its book value R118 636,75? (3)

Asked: 2020 (reducing balance)

7.1.5 A printer bought for R15 000 depreciates at 12,5% p.a. on the straight-line method. After how many years will it have a book value of R0? (2)

Asked: 2023

7.1.6 After 5 years the value of a machine was 40% of its original value. Determine the annual rate of depreciation if the straight-line method was used. (2)

Asked: 2024

7.2.1 R14 000 was invested for 3 years at an interest rate of $r\%$ p.a., compounded monthly. At the end of this period the investment was worth R17 364,22. Calculate r, correct to ONE decimal place. (5)

Asked: 2017, 2023

7.2.2 Use your answer to QUESTION 7.2.1 to calculate the effective annual interest rate. (2)

Asked: 2023

7.2.3 R9 500 was invested in a fund that paid interest at $m\%$ p.a., compounded quarterly. After 15 months the value of the investment was R10 285. Determine the value of m. (4)

Asked: 2022 (time in months, compounded quarterly)

7.2.4 How many years will it take for R60 000 to accumulate to R88 838,74 at 7,2% p.a., compounded quarterly? (4)

Asked: 2021

7.2.5 Zinzi and Ben each want to invest R8 000 for 5 years. Zinzi invests in an account that pays simple interest at 9,1% p.a. At the end of 5 years she receives a bonus of exactly 3% of the accumulated amount. Ben invests in an account that pays 8,4% p.a., compounded quarterly. Whose investment will be worth more at the end of 5 years? Justify your answer with appropriate calculations. (5)

Asked: 2019

On 31 March 2021, Palesa made the first of her monthly deposits of R750 into a savings account. She continues to make deposits of R750 at the end of each month up to and including 31 March 2030. The interest rate was fixed at 8,4% p.a., compounded monthly.

7.3.1 What will the investment be worth immediately after her last deposit? (4)

Asked: 2020

7.3.2 If she makes no further deposits but leaves the money in the account, how much will be in the account on 31 March 2031? (2)

Asked: 2020

7.3.3 Zola deposited R1 200 into an account on 31 January 2024 and continued to deposit R1 200 on the last day of every month. Her last deposit will be on 31 December 2024. The account pays 7,2% p.a., compounded monthly. Will she have enough money in the account on 1 January 2025 to buy a laptop that costs R15 500? Justify your answer with an appropriate calculation. (4)

Asked: 2022

7.3.4 Themba wants to save R380 000 to replace a delivery van. He makes a fixed deposit at the end of every month into an account that earns 9,3% p.a., compounded monthly. His first deposit is made one month from now and his 60th deposit is made when the van is to be replaced. Calculate his monthly deposit. (4)

Asked: 2021, 2023 (sinking fund)

7.3.5 Karabo opened a savings account that pays interest at 6,4% p.a., compounded quarterly. She deposited R1 800 into the account on 1 January 2021 and continued to deposit R1 800 at the BEGINNING of each quarter after that. Her last deposit was on 1 October 2026. Calculate the accumulated amount in the account on 1 January 2027. (4)

Asked: 2025

Selby decided today that he will save R12 000 per quarter for the next four years. He will make the first deposit into a savings account in three months' time and his last deposit at the end of four years from now. Interest is earned at 7,6% p.a., compounded quarterly.

7.3.6 How much will Selby have at the end of four years? (3)

Asked: 2018

7.3.7 Selby withdraws R80 000 from the account at the end of three years from now. How much will he have in the account at the end of four years from now? (3)

Asked: 2018

On 1 March 2021 a bank granted Bheki a loan of R320 000 at 13,2% p.a., compounded monthly. He agreed to repay the loan in monthly instalments, starting on 1 April 2021 and ending five years later on 1 March 2026. However, Bheki was unable to pay the first three instalments and only started paying on 1 July 2021.

7.4.1 Calculate the amount Bheki owed the bank on 1 June 2021, a month before he paid his first instalment. (2)

Asked: 2016

7.4.2 Bheki will still pay his last instalment on 1 March 2026. Calculate his monthly instalment. (4)

Asked: 2016

7.4.3 Bheki decides to pay R9 000 per month, starting on 1 July 2021. How many months sooner than 1 March 2026 will he settle the loan? (5)

Asked: 2016

7.4.4 With payments of R9 000 per month, calculate the final (smaller) instalment needed to settle the loan. (4)

Asked: 2016, 2025

Pieter takes a loan of R184 000 from a bank to buy a car. He agrees to repay the loan over 48 months. The first instalment is paid one month after the loan is granted. The bank charges interest at 12,6% p.a., compounded monthly.

7.5.1 Calculate Pieter's monthly instalment. (4)
Asked: 2017

7.5.2 Calculate the total amount of interest that Pieter will pay during the first year of the repayment of the loan. (6)
Asked: 2017, 2020, 2021

Tumi takes out a home loan of R1 350 000 over 20 years. Interest is charged at 9,9% p.a., compounded monthly. The first instalment is paid one month after the loan is granted.

7.6.1 Calculate the monthly instalment. (4)
Asked: 2018

7.6.2 Calculate the outstanding balance immediately after the 150th payment was made. (5)
Asked: 2018

Mpho is granted a loan under these conditions: the interest rate is 10,9% p.a., compounded monthly; the period of the loan is 7 years; the monthly repayment is R1 800; her first repayment is made one month after the loan is granted.

7.7 Calculate the value of the loan. (3)
Asked: 2020

Sam buys a house costing R1 380 000 and pays a deposit of R180 000. He gets a home loan for the balance at 11,4% p.a., compounded monthly, over 20 years. The loan was granted on 1 January 2019 and he paid his first monthly instalment of R12 714,57 on 31 January 2019.

7.8 What will the balance outstanding on the loan be on 31 December 2035, which is 17 years after the loan was granted? (3)
Asked: 2021

Thabo plans to buy a car that costs R340 000. He will pay a deposit of 20% and take out a loan for the balance. The interest on the loan is 12,6% p.a., compounded monthly.

7.9.1 Calculate the value of the loan. (1)
Asked: 2022

7.9.2 The first repayment will be made 5 months after the loan has been granted. The loan will be repaid over a period of 5 years after it has been granted. Calculate the MONTHLY instalment. (5)
Asked: 2022

Tshepo was granted a loan of R150 000 on 1 August 2021 at 14,4% p.a., compounded monthly. He agreed to repay the loan over 6 years in monthly instalments of R3 123,08, starting on 1 September 2021.

7.10.1 Calculate the total interest that he will pay over the 6 years. (2)
Asked: 2024

7.10.2 On 1 August 2024, Tshepo paid R28 123,08 (his monthly instalment and an additional R25 000) into the loan account. He continues to pay the original monthly instalment thereafter. How many months earlier than planned will Tshepo repay the loan? (7)
Asked: 2024

Eight years ago a bank granted Mandy a home loan of R640 000. The loan was to be repaid over 20 years at 9,6% p.a., compounded monthly. Her monthly repayments started exactly one month after the loan was granted. Mandy decided to repay R7 200 per month instead of the required R6 007,49.

7.11.1 How many payments will she make to settle the loan, counting from the day the loan was granted? (5)

Asked: 2019

7.11.2 After repaying R7 200 per month for 8 years, Mandy needs money for her son's studies. The bank tells her to regard the extra amount she paid each month as a savings plan and that she may withdraw the whole amount. Calculate the maximum amount that Mandy may withdraw from the loan account. (4)

Asked: 2019

7.12 Tino wins a jackpot of R1 200 000. He invests all of it in a fund that earns 10,2% p.a., compounded monthly. He withdraws R15 000 from the fund at the end of each month, the first withdrawal being exactly one month after the investment. How many withdrawals of this amount can Tino make? (5)

Asked: 2023

A bank granted Rajesh a loan of R680 000 on 31 January 2023 at 8,4% p.a., compounded monthly. He was supposed to start repaying on 28 February 2023 but could not make the first three payments. He made his first repayment of R9 500 on 31 May 2023 and continued to repay R9 500 at the end of each month after that.

7.13 How long, in completed months from the day the loan was granted, will it take Rajesh to repay the loan? (5)

Asked: 2025

[133]

QUESTION 8: DIFFERENTIAL CALCULUS: FIRST PRINCIPLES AND RULES [89 marks of variants]

Every year: the derivative from first principles (4 to 6 marks, limit notation on every line) and several derivatives by the rules (power rule with surds and negative powers, sometimes after simplifying first). Often: a tangent equation found with the rules. Sometimes: a limit written as a derivative, partial derivatives, minimum gradient with concavity, or an inverse function. 10 to 18 marks.

First principles. Use the definition $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$.

8.1.1 Determine $f'(x)$ from first principles if $f(x) = 3x^2 - 4$. (5)

Asked: 2018, 2020 (a quadratic with a constant term)

8.1.2 Determine $f'(x)$ from first principles if it is given that $f(x) = 3x^2 - 2x$. (6)

Asked: 2017 (6 marks), 2021, 2022

8.1.3 Determine $f'(x)$ from first principles if it is given that $f(x) = 8 - 5x$. (4)

Asked: 2019, 2025 (a linear function)

8.1.4 Thabo determines $g'(a)$, the derivative of a function g at $x = a$, and arrives at the answer $\lim_{h \to 0} \dfrac{\dfrac{1}{3+h} - \dfrac{1}{3}}{h}$. Write down the equation of g and the value of a. (2)

Asked: 2016

Differentiation rules. Do NOT use first principles in QUESTION 8.2.

8.2.1 Determine $\dfrac{dy}{dx}$ if $y = 2x^5 - 7x^3 + 4x$. (3)

Asked: 2018, 2021, 2022 (written as $f'(x)$), 2023 and 2025 (the same, with two terms)

8.2.2 Determine: $\dfrac{d}{dx}\left[9x - 4x^2\right]$ (2)

Asked: 2024 (notation $\frac{d}{dx}$)

8.2.3 Determine: $D_x\left[(x+2)(4x-5)\right]$ (2)

Asked: 2017 (expand, then differentiate)

8.2.4 Determine $\dfrac{dy}{dx}$ if $y = \sqrt{x^5} - \dfrac{6}{x^2}$. (4)

Asked: 2016

8.2.5 Determine $\dfrac{dy}{dx}$ if $y = \dfrac{4}{x} + \sqrt{x^3} - \dfrac{1}{3}\pi$. (4)

Asked: 2017 (a constant π term)

8.2.6 Determine $\dfrac{dy}{dx}$ if $y = 3x^6 + \sqrt[4]{x^3}$. (3)

Asked: 2019, 2020

8.2.7 Determine: $D_x\left[-\dfrac{\sqrt{x}}{4} + \left(\dfrac{1}{2x}\right)^3\right]$ (4)

Asked: 2021

8.2.8 Determine: $D_x\left(10\sqrt[5]{x^3} + 4x^{-3}\right)$ (3)

Asked: 2023

8.2.9 Determine $g'(x)$ if $g(x) = \dfrac{3}{x^4} - \sqrt[3]{x^5}$. (4)

Asked: 2024

8.2.10 Determine $\frac{dy}{dx}$ if $yx - 2y = 3x^2 - 6x$; $x \neq 2$. (4)

Asked: 2018 (make y the subject first)

8.2.11 Determine $f'(x)$ if $f(x) = \frac{25x^2 - 4}{10x - 4}$; $x \neq \frac{2}{5}$. (4)

Asked: 2020 (factorise, cancel, differentiate)

8.2.12 Determine $\frac{dy}{dx}$ if $y = \frac{4x^5 + 3}{x^2}$. (4)

Asked: 2025 (split the fraction)

Partial derivatives. Given: $y = ax^3 - 5a$, where a is a constant.

8.3.1 Determine $\frac{dy}{dx}$. (1)

Asked: 2019

8.3.2 Determine $\frac{dy}{da}$. (2)

Asked: 2019

Tangents.

8.4.1 $g(x) = -5x + 8$ is a tangent to $f(x) = x^3 + ax^2 + bx + 12$ at $x = 2$. Calculate the values of a and b. (5)

Asked: 2016

8.4.2 The curve with equation $y = x + \frac{8}{x}$ passes through the point $A(4; b)$. Determine the equation of the line perpendicular to the tangent to the curve at A. (4)

Asked: 2019 (perpendicular to the tangent)

8.4.3 Determine the equation of the tangent to $f(x) = x^3 - 5x^2 + 4x + 2$ at $x = 3$. (3)

Asked: 2024

Gradients of tangents and concavity.

8.5.1 For which values of x will the tangent to $f(x) = -x^3 + 3x^2 + 9x$ have a positive gradient? (3)

Asked: 2023

8.5.2 The tangent to $g(x) = ax^3 + 6x^2 + bx + c$ has a minimum gradient at the point $(-2; 5)$. For which values of x will g be concave up? (4)

Asked: 2022

An inverse function. Given: $f(x) = -3x^2$.

8.6.1 Determine $f'(x)$ from first principles. (5)

Asked: 2016, 2023, 2024 (the form ax^2)

8.6.2 Write down how you will restrict the domain of f so that f^{-1}, the inverse of f, is a function. (1)

Asked: 2024

8.6.3 Determine the equation of f^{-1} for $f^{-1}(x) \geq 0$. Write your answer in the form $y = \ldots$ (3)

Asked: 2024

[89]

QUESTION 9: DIFFERENTIAL CALCULUS: CUBIC GRAPHS AND TANGENTS [146 marks of variants]

Every year: a cubic function (given by its equation, its derivative, its graph or a few conditions). Always: turning points and concavity, usually the point of inflection, increasing or decreasing intervals and a sketch. Often: finding the equation from the graph, the values of k for which $f(x) = k$ has a given number of roots, a tangent at a special point, or the sign of a product such as $f(x) \cdot f''(x)$. 8 to 18 marks.

For a certain function f, the first derivative is given as $f'(x) = 6x^2 - 6x - 12$.

9.1.1 Calculate the x-coordinates of the stationary points of f. (3)

Asked: 2016

9.1.2 For which values of x is f concave down? (3)

Asked: 2016

9.1.3 Determine the values of x for which f is strictly increasing. (2)

Asked: 2016

9.1.4 If it is further given that $f(x) = ax^3 + bx^2 + cx + d$ and $f(0) = 5$, determine the equation of f. (5)

Asked: 2016

Given: $f(x) = (x + 1)(x - 2)^2$, with $f'(0) = f'(2) = 0$ and $f(0) = 4$.

9.2.1 Show that f has a point of inflection at $x = 1$. (5)

Asked: 2017

9.2.2 Sketch the graph of f. Clearly indicate the intercepts with the axes and the turning points. (4)

Asked: 2017

9.2.3 For which values of x will $y = -f(x)$ be concave down? (2)

Asked: 2017

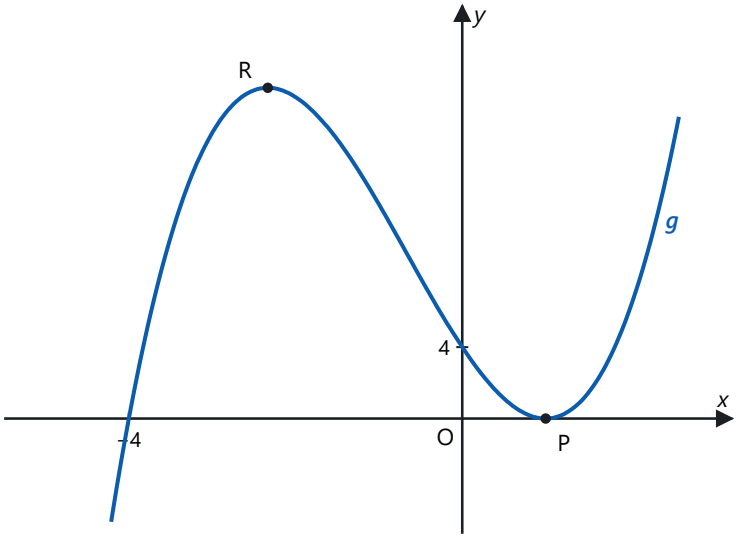
9.2.4 Use your graph to determine the coordinates of the local maximum of h if $h(x) = f(x + 2) - 3$. (2)

Asked: 2017

9.2.5 Sipho claims that $f'(1) = 3$. Do you agree with Sipho? Justify your answer. (2)

Asked: 2017

The graph of $g(x) = x^3 + bx^2 + cx + d$ is sketched below. The graph of g intersects the x-axis at $(-4; 0)$ and at P, and the y-axis at $(0; 4)$. P and R are turning points of g.



9.3.1 Show that $b = 2$, $c = -7$ and $d = 4$. (4)

Asked: 2018

9.3.2 Calculate the coordinates of P and R. (5)

Asked: 2018

9.3.3 Is the graph of g concave up or concave down at $(0; 4)$? Show ALL your calculations. (3)

Asked: 2018

9.3.4 If p is a cubic function with $p(2) = p'(2) = 0$, $p(0) = -8$, $p''(x) < 0$ when $x < 2$ and $p''(x) > 0$ when $x > 2$, draw a sketch graph of p. Indicate ALL relevant points. (3)

Asked: 2018

Given: $f(x) = 2x^3$. The graphs of f, f' and f'' all pass through the point $(0; 0)$.

9.4.1 Solve $f(x) = f'(x)$. (3)

Asked: 2019

9.4.2 For which of the graphs will $(0; 0)$ be a stationary point? (1)

Asked: 2019

9.4.3 Explain the difference, if any, in the stationary points referred to in QUESTION 9.4.2. (2)

Asked: 2019

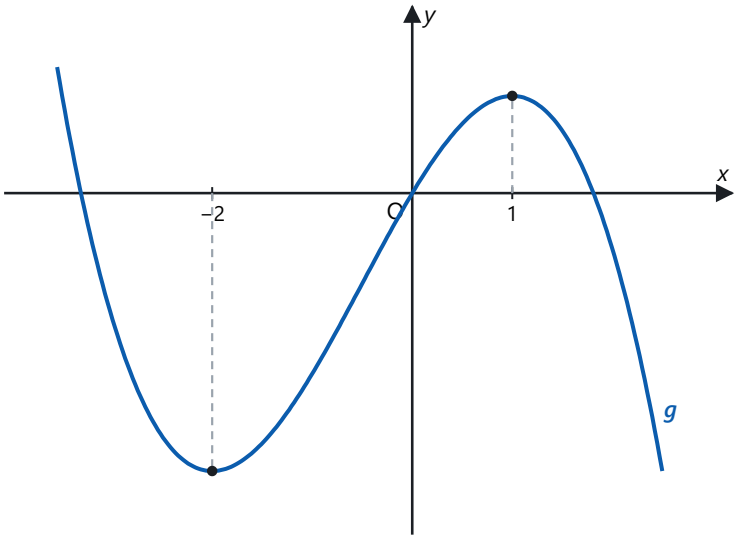
9.4.4 Determine the vertical distance between the graphs of f' and f'' at $x = 1$. (3)

Asked: 2019

9.4.5 For which value(s) of x is $f(x) - f'(x) < 0$? (4)

Asked: 2019

The graph of $g(x) = ax^3 + bx^2 + cx$, a cubic function with a y-intercept of 0, is drawn below. The x-coordinates of the turning points of g are -2 and 1.



9.5.1 For which values of x will g increase? (2)

Asked: 2020

9.5.2 Write down the x-coordinate of the point of inflection of g. (2)

Asked: 2020

9.5.3 For which values of x will g be concave down? (2)

Asked: 2020

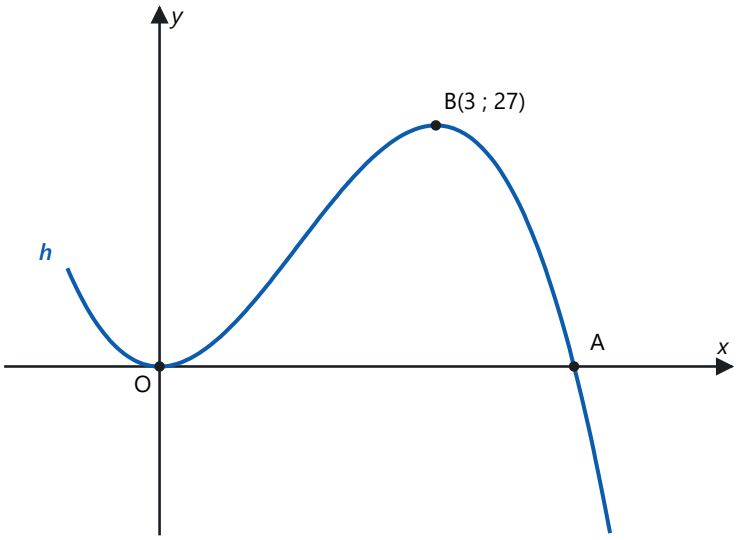
9.5.4 If $g'(x) = -6x^2 - 6x + 12$, determine the equation of g. (4)

Asked: 2020

9.5.5 Determine the equation of the tangent to g that has the maximum gradient. Write your answer in the form $y = mx + c$. (5)

Asked: 2020

The graph of $h(x) = ax^3 + bx^2$ is drawn below. The graph has turning points at the origin $O(0;0)$ and at $B(3;27)$. A is an x-intercept of h.



9.6.1 Show that $a = -2$ and $b = 9$. (5)

Asked: 2021

9.6.2 Calculate the coordinates of A. (3)

Asked: 2021

9.6.3 Write down the values of x for which h is increasing. (2)

Asked: 2021

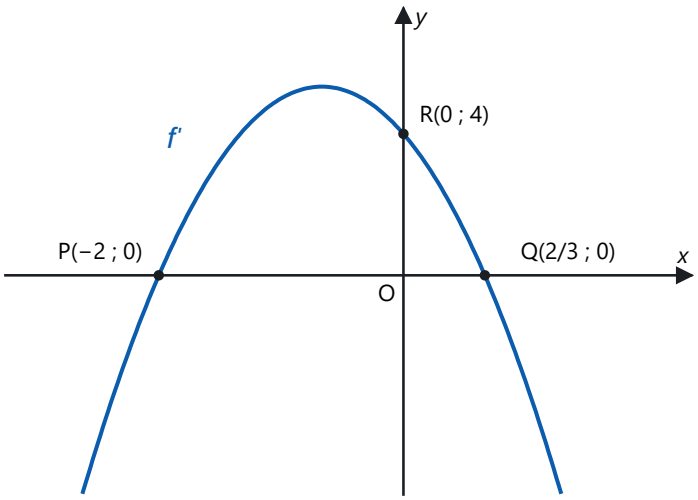
9.6.4 Write down the values of x for which h is concave down. (2)

Asked: 2021

9.6.5 For which values of k will $-2(x+2)^3 + 9(x+2)^2 - k = 0$ have one negative and two distinct positive roots? (3)

Asked: 2021

The graph of $y = f'(x) = mx^2 + nx + k$ is drawn below. The graph passes through the points $P(-2;0)$, $Q(\frac{2}{3};0)$ and $R(0;4)$.



9.7.1 Determine the values of m, n and k. (6)

Asked: 2022

It is further given that $f(x) = -x^3 - 2x^2 + 4x + 8$.

9.7.2 Determine the coordinates of the turning points of f. (3)

Asked: 2022

9.7.3 Draw the graph of f. Indicate on your graph the coordinates of the turning points and the intercepts with the axes. (5)

Asked: 2022

Points E and W are two variable points on f' and are on the same horizontal line. h is a tangent to f' at E and g is a tangent to f' at W. h and g intersect at $D(a; b)$.

9.7.4 Write down the value of a. (1)

Asked: 2022

9.7.5 Determine the value(s) of b for which h and g will no longer be tangents to f'. (2)

Asked: 2022

Given: $f(x) = -2x^3 + 12x^2 - 18x = -2x(x-3)^2$.

9.8.1 Determine the coordinates of the turning points of f. (4)

Asked: 2023

9.8.2 Draw a sketch graph of f. Clearly label all the intercepts with the axes and any turning points. (4)

Asked: 2023

9.8.3 Use the graph to determine the value(s) of k for which $-2x^3 + 12x^2 - 18x = k$ will have three real and unequal roots. (2)

Asked: 2023

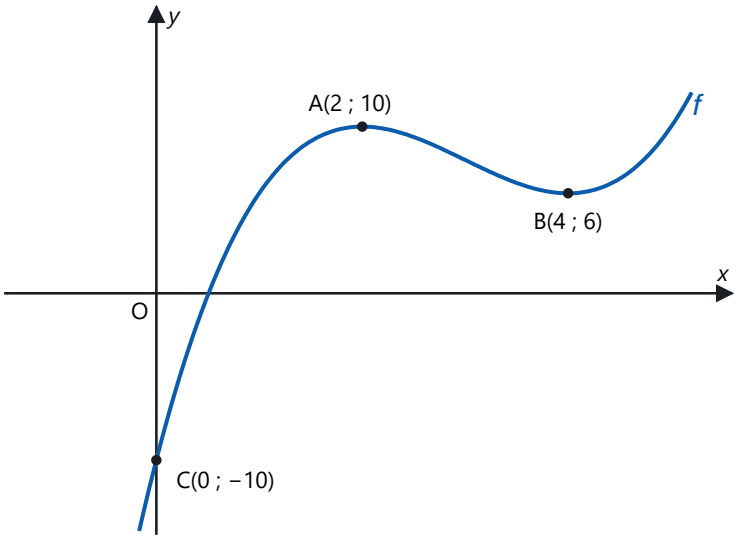
9.8.4 The line $g(x) = ax + b$ is the tangent to f at the point of inflection of f. Determine the equation of g. (6)

Asked: 2023

9.8.5 Calculate the value of θ, the acute angle formed between g and the x-axis in the first quadrant. (2)

Asked: 2023

$A(2; 10)$ and $B(4; 6)$ are the turning points of the graph of f below. $C(0; -10)$ is the y-intercept of f.



9.9.1 For which values of x is f decreasing? (2)

Asked: 2024

9.9.2 Write down the x-intercepts of f', the derivative of f. (2)

Asked: 2024

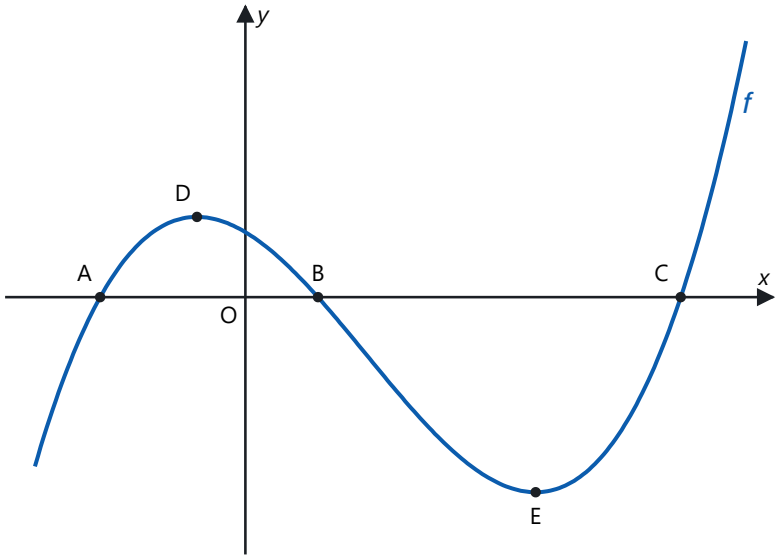
9.9.3 For which values of x will f be concave up? (2)

Asked: 2024

9.9.4 Determine the values of k for which $y = f(x) + k$ will have THREE positive x-intercepts. (2)

Asked: 2024

Sketched below is the graph of $f(x) = x^3 - 5x^2 - 8x + 12$. A, B and C are the x-intercepts of f. D and E are the turning points of f.



9.10.1 Calculate the coordinates of E. (4)

Asked: 2025

9.10.2 For which values of x is f concave down? (3)

Asked: 2025

9.10.3 The coordinates of B are $(1; 0)$. Use the graph to determine the values of x for which $f(x) \cdot f''(x) < 0$. (4)

Asked: 2025

9.10.4 For which values of t will the line $y = -8x + t$ intersect f at 3 distinct points? (6)

Asked: 2025

[146]

QUESTION 10: CALCULUS: OPTIMISATION AND RATES OF CHANGE [74 marks of variants]

Every year: one or two applications of differentiation in a context. Always: set up a function (often from a given diagram or a cost, area or volume story), differentiate it and equate the derivative to zero to find the maximum or minimum. Often: 'show that' the function first, then optimise it. Sometimes: the rate of change at a given moment, the time of a maximum rate (second derivative), a given function to evaluate, or a speed function you work back from (find s(t) from s'(t)) to get a distance. 6–10 marks per context.

The number of bacteria (in thousands) in a culture t hours after a nutrient is added is given by $N(t) = -t^3 + 6t^2 + 36t$, for $0 < t < 9$.

10.1.1 Determine the number of bacteria in the culture 4 hours after the nutrient was added. (2)

Asked: 2016

10.1.2 Determine the rate at which the number of bacteria is changing at exactly 3 hours after the nutrient was added. (3)

Asked: 2016

10.1.3 How many hours after the nutrient was added will the rate at which the number of bacteria is changing be a maximum? (3)

Asked: 2016

A gecko crawls up a wall. The height of the gecko above the floor is described by $h(t) = (t - 8)(-2t^2 + 2t - 14)$, where h is the height in cm and t is the time in minutes since the gecko started crawling.

10.2.1 At what height above the floor did the gecko start to crawl? (1)

Asked: 2019

10.2.2 How many times did the gecko reach the floor? (3)

Asked: 2019

10.2.3 Determine the maximum height that the gecko reached above the floor. (4)

Asked: 2019

A train leaves station P and stops at station T. The speed of the train (in km/h) is given by $s'(t) = -3t^2 + 30t$, where t is the time in hours since the train left P. **NOTE:** Speed is the rate of change in distance with respect to time.

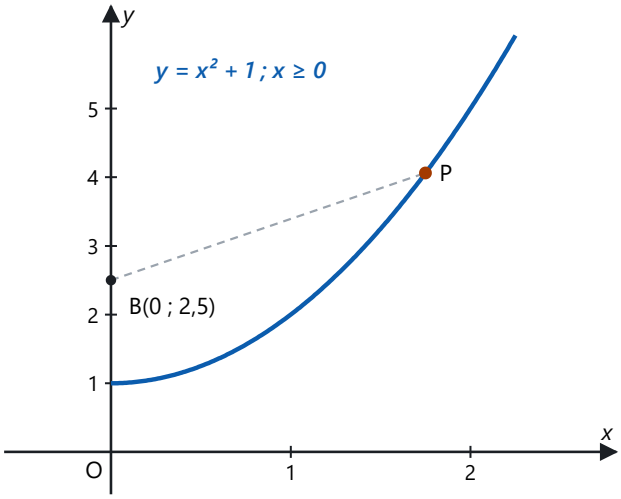
10.3.1 Calculate the maximum speed that the train reached on this journey. (3)

Asked: 2024

10.3.2 Calculate the distance between station P and station T. (5)

Asked: 2024

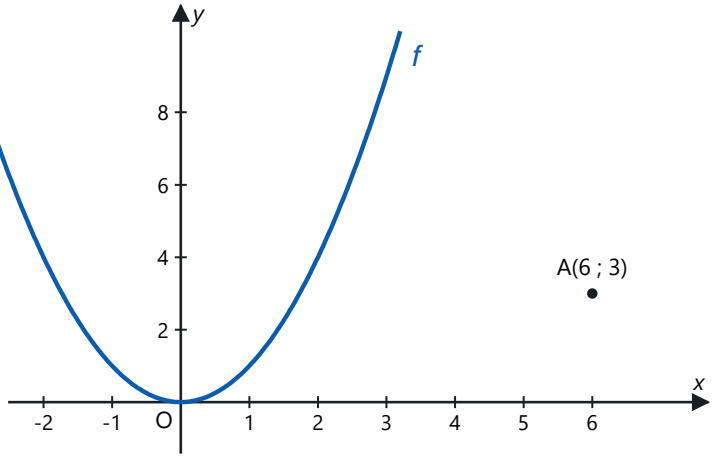
The road in the diagram below is described by $y = x^2 + 1$, for $x \geq 0$, where the axes (dotted in the aerial view) are chosen as shown. Beth sits at a vantage point $B(0; 2{,}5)$ and watches a car, P, travelling along the road.



10.4 Calculate the distance between Beth and the car when the car is closest to Beth. (7)

Asked: 2017

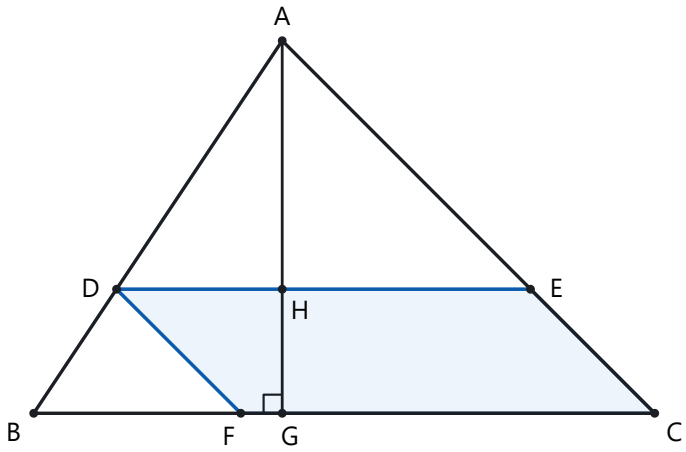
Given: $f(x) = x^2$. The point $A(6; 3)$ is shown on the diagram.



10.5 Determine the minimum distance between the point $A(6; 3)$ and a point on f. (8)

Asked: 2022

In $\triangle ABC$, D is a point on AB, E is a point on AC and F is a point on BC such that DECF is a parallelogram. $BF : FC = 1 : 2$. The perpendicular height AG is drawn, intersecting DE at H. $AG = t$ units and $BC = (6 - t)$ units.



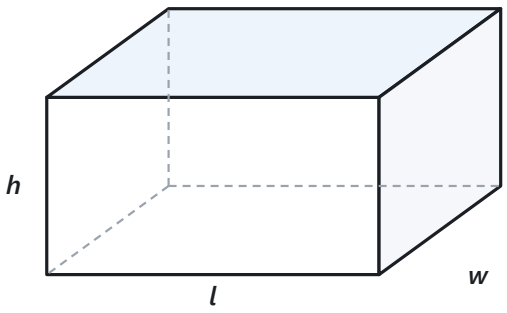
10.6.1 Write down the ratio $AH : HG$. (1)

Asked: 2018

10.6.2 Calculate t if the area of the parallelogram is a maximum. (NOTE: Area of a parallelogram = base $\times$ perpendicular height.) (5)

Asked: 2018

A closed rectangular storage chest has to be constructed as follows: the dimensions are length l, width w and height h (all in metres); the length l of the base has to be twice its width w; the volume has to be $32\ \text{m}^3$. The material for the top and the bottom costs R12 per square metre and the material for the sides costs R8 per square metre.



10.7.1 Show that the cost to construct the chest can be calculated by: $\text{Cost} = 48w^2 + 48wh$ (4)

Asked: 2020

10.7.2 Determine the width of the chest such that the cost to build the chest is a minimum. (6)

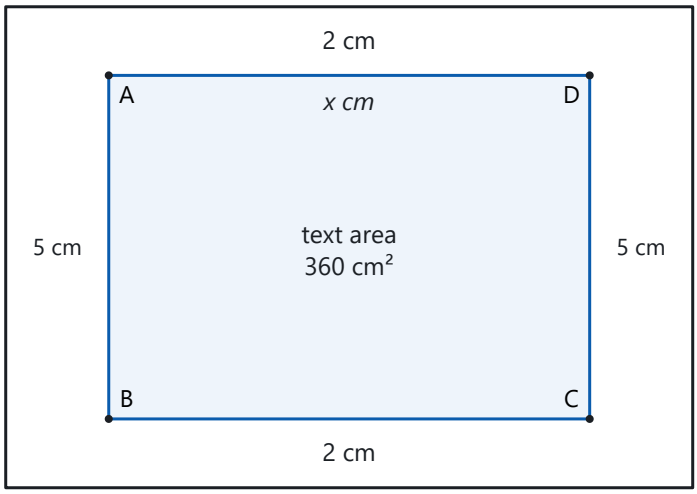
Asked: 2020

After driving 150 km from the depot, a delivery driver is called back and returns to the depot immediately. The driver is paid R40 per hour for the time of the trip. The cost of petrol is $\left(1{,}4 + \frac{x}{250}\right)$ rands per km, where x is the average speed in km/h.

10.8 Calculate the average speed at which the driver must travel back to keep the total cost as low as possible. (7)

Asked: 2021

The diagram shows a printed notice. The rectangle ABCD is the part on which the text is printed. The text area ABCD is $360\ \text{cm}^2$ and $AD = x$ cm. ABCD is 5 cm from the left and the right edges of the notice and 2 cm from the top and the bottom edges.



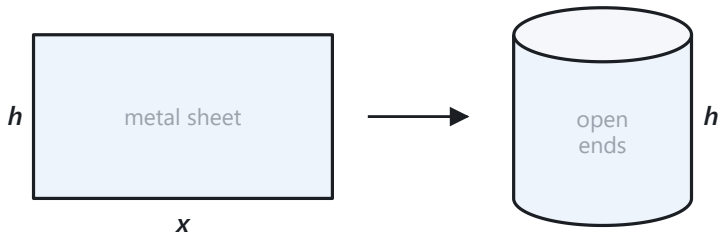
10.9.1 Show that the total area of the notice is given by: $A(x) = \dfrac{3600}{x} + 4x + 400$ (3)

Asked: 2023

10.9.2 Determine the value of x such that the total area of the notice is a minimum. (3)

Asked: 2023

A rectangular metal sheet has dimensions x and h units, with $x > h$, and a perimeter of 36 units. The sheet is rolled into a cylinder with two open ends (top and bottom). The side of length x becomes the circumference of the cylinder and the cylinder has height h units.



10.10.1 Show that the volume of the cylinder is given by: $V = \dfrac{18x^2 - x^3}{4\pi}$ (3)

Asked: 2025

10.10.2 Calculate the value of x that will maximise the volume of the cylinder. (3)

Asked: 2025

[74]

QUESTION 11: PROBABILITY

[88 marks of variants]

Every year: events A and B (mutually exclusive or independent), a Venn diagram, a two-way table or a tree diagram, with probabilities found from them. Always: a test for independence or mutually exclusive events, with the working shown. Usually: a Venn diagram with two or three events (often drawn by the learner), a probability that uses 'at least', 'only' or 'not'. Sometimes: an unknown found from an equation, repeated shots with a geometric series, or a two-draw problem. 3–9 marks per part.

Given: $P(A) = 0{,}4$; $P(B) = y$ and $P(A \text{ or } B) = 0{,}7$

11.1.1 Determine the value of y if A and B are mutually exclusive. (3)

Asked: 2018

Events A and B are independent. $P(A) = 0{,}6$ and $P(B) = 0{,}35$.

11.2.1 Represent the given information on a Venn diagram. Indicate on the Venn diagram the probabilities associated with each region. (3)

Asked: 2019

11.2.2 Determine $P(A \text{ or NOT } B)$. (2)

Asked: 2019

A and B are independent events. It is further given that $P(A \text{ and } B) = 0{,}24$ and $P(\text{only } B) = 0{,}16$.

11.3.1 Are A and B mutually exclusive? Motivate your answer. (1)

Asked: 2021

11.3.2 Determine $P(\text{only } A)$. (4)

Asked: 2021

11.3.3 Determine $P(\text{not } A \text{ or not } B)$. (2)

Asked: 2021

A and B are independent events. $P(A) = \frac{2}{5}$ and $P(B) = \frac{5}{8}$.

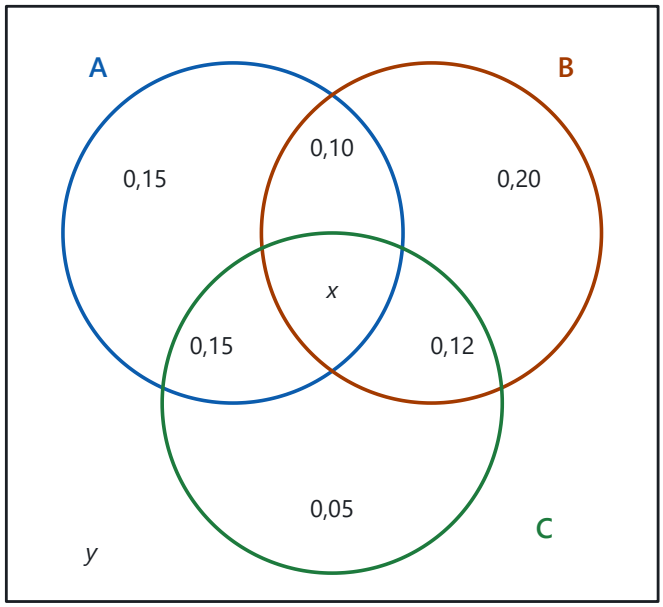
11.4.1 Determine $P(A \text{ and } B)$. (2)

Asked: 2023

11.4.2 Determine $P(\text{at least ONE event occurs})$. (2)

Asked: 2023

A, B and C are three events. The probabilities of these events (or any combination of them) occurring are given in the Venn diagram below.



11.5.1 If it is given that the probability that at least one of the events will occur is 0,85, calculate the value of y, the probability that none of the events will occur. (1)

Asked: 2022

11.5.2 With the same information, calculate the value of x, the probability that all three events will occur. (1)

Asked: 2022

11.5.3 Determine the probability that at least TWO of the events will take place. (2)

Asked: 2022

11.5.4 Are the events B and C independent? Justify your answer. (5)

Asked: 2022

A survey was conducted among 100 Grade 12 learners about the sports they play: soccer (S), cricket (C) and hockey (H). The survey revealed the following:
6 play all three sports.
10 play soccer and cricket.
4 play cricket and hockey, but not soccer.
x play soccer and hockey, but not cricket.
40 play soccer.
22 play cricket.
35 play hockey.
32 play none of these sports.

11.6.1 Draw a Venn diagram to illustrate the information above. (4)

Asked: 2017

11.6.2 Calculate the value of x. (2)

Asked: 2017

11.6.3 Calculate the probability that a learner, chosen randomly, plays only ONE of these sports. (2)

Asked: 2017

A group of learners belong to at least one of three school clubs: Chess (C), Drama (D) and Debating (B). The total number of learners in Chess is 21. The total number in Drama is 19. The total number in Debating is 19. 6 learners belong to Chess and Drama, but not Debating. 5 learners belong to Chess and Debating, but not Drama. 4 learners belong to Drama and Debating, but not Chess. 7 learners belong to Drama only.

11.7.1 Draw a Venn diagram to represent ALL the learners in these clubs. (3)

Asked: 2024

11.7.2 Calculate the probability that a learner, chosen at random, belongs to at least TWO of the clubs. (2)

Asked: 2024

11.7.3 Determine if belonging to Chess and belonging to Drama are independent events. Support your answer with the necessary calculations. (4)

Asked: 2024

A survey was conducted among 90 boys and 70 girls to find out whether they play a sport. Their responses are shown in the partially completed table below.

	Plays a sport	Does not play a sport	Totals
Boys	72	a	
Girls	48	22	
Totals	b	40	160

11.8.1 Calculate the values of a and b. (2)

Asked: 2016

11.8.2 Are the events 'being a boy' and 'does not play a sport' mutually exclusive? Give a reason for your answer. (2)

Asked: 2016

11.8.3 If a learner who took part in this survey is chosen at random, what is the probability that the learner plays a sport? (2)

Asked: 2016

11.8.4 What is the probability that the learner is not a boy and does not play a sport? (2)

Asked: 2016

A survey was conducted among 250 learners at a school about how they travel to school. The data from the survey is presented in the table below.

	Walk	Taxi	Total
Female	a	b	c
Male	40	60	f
Total	e	d	250

11.9.1 The events 'male' and 'walks to school' are independent. Show that $e = 100$. (3)

Asked: 2025

11.9.2 Calculate the probability that a female learner, chosen at random from the group, travels to school by taxi. (3)

Asked: 2025

The probability that it will rain on the day of a school netball match is 30%. If it rains, the probability that the home team wins is 40%. If it does not rain, the probability that the home team wins is 65%.

11.10.1 Represent the given information on a tree diagram. Clearly indicate the probabilities associated with EACH branch. (3)

Asked: 2023

11.10.2 Calculate the probability that the home team does NOT win. (3)

Asked: 2023

A café sells either iced tea or coffee to each of its 160 customers on a day. The chance that a day is hot is 60%. The chance that a customer buys iced tea on a hot day is twice the chance that the customer buys iced tea on a day that is not hot. The probability that a customer buys iced tea on any given day is $\frac{2}{5}$.

11.11 Calculate the number of customers who will buy iced tea on a day that is not hot. (4)

Asked: 2025

The school tuck shop is open from Monday to Friday. Luthando and Mia each bought lunch at the tuck shop on one day this week. If any day is equally likely, calculate the probability that they bought lunch on:

11.12.1 The same day (2)

Asked: 2019

11.12.2 Consecutive days (3)

Asked: 2019

Sipho shoots arrows at a target board. He has a 40% chance of hitting the bull's eye on each shot.

11.13.1 Calculate the probability that Sipho will hit the bull's eye with his first shot and his second shot. (2)

Asked: 2020

11.13.2 Calculate the probability that Sipho will hit the bull's eye at least twice in his first three shots. (3)

Asked: 2020

11.13.3 Thandi also has a 40% chance of hitting the bull's eye on each shot. Sipho and Thandi take turns to shoot an arrow, and the first person to hit the bull's eye wins. Sipho shoots first. Calculate the probability that Sipho wins. (3)

Asked: 2020

A shop gives out gift bags of sweets to a Grade R class. There is a mystery gift in exactly $\frac{1}{3}$ of the total number of bags. Each learner randomly selects two gift bags, one after the other.

11.14 The probability that a learner selects two bags that both have a mystery gift is $\frac{4}{39}$. Calculate the number of bags that have a mystery gift inside. (6)

Asked: 2018

[88]

QUESTION 12: COUNTING PRINCIPLE [65 marks of variants]

Every year: use the fundamental counting principle to count codes, numbers or arrangements, usually with a restriction such as even, odd, greater than a given value, or no repeats. Often: the restricted count is used to find a probability, or a count with and without a restriction is compared as a percentage increase. Sometimes: a context such as a car showroom, books on a shelf, a line of learners or a race. 5–17 marks, often spread over several parts.

The digits 1 to 8 are used to create a four-digit code to open a locked cupboard.

12.1 How many different codes are possible if the digits may not be repeated and the code must be an odd number bigger than 6 000? (5)

Asked: 2016 (even code, greater than 5 000)

A club gives each member a code made up of two letters followed by a sequence of digits, for example KT4087. The letters are chosen from B ; K ; M ; P ; T and Z. Letters may be repeated in the code. The digits 0 to 9 are used, but NO digit may be repeated in the code.

12.2.1 How many different members can be identified with a code made up of TWO letters and TWO digits? (3)

Asked: 2017

12.2.2 Determine the least number of digits that is required to identify 1 000 000 members using this coding system. (3)

Asked: 2017

Given the digits: 2 ; 3 ; 4 ; 5 ; 6 ; 7 ; 8 and 9

12.3.1 Calculate how many unique 4-digit codes can be formed using the digits above if the digits may be repeated. (2)

Asked: 2018

12.3.2 Calculate how many unique 4-digit codes can be formed using the digits above if the digits may NOT be repeated. (2)

Asked: 2018

12.3.3 How many unique 3-digit codes can be formed using the digits above if digits may be repeated, the code is greater than 300 but less than 700, and the code is divisible by 5? (3)

Asked: 2018

A sports club issues 9-digit membership numbers in the format shown below. Digits may be repeated.

Format	Region code	Branch code	Serial number
Number of digits	2 digits	3 digits	4 digits
Example	47	380	5620

12.4.1 How many possible 9-digit membership numbers could be formed? (2)

Asked: 2020

12.4.2 Restrictions were placed on the groups of digits. (3)
Region code: the first digit can NOT be 0.
Branch code: the first and the last digits can NOT be 0 or 9.
Serial number: the last digit MUST be 0 or 5.
How many valid membership numbers could be formed by applying these restrictions?

Asked: 2020

12.4.3 Determine the probability that a 9-digit number chosen at random would be a valid membership number. (2)

Asked: 2020

A company generates a 4-character code using the 26 letters of the alphabet and the 10 digits from 0 to 9. The code is in the form shown below.

letter	letter	digit	digit

12.5.1 Determine how many different codes can be formed if letters and digits may be repeated. (2)

Asked: 2024

12.5.2 Determine how many different codes can be formed if: (4)

- the letters B, C, D and F may NOT be used
- the code may NOT start with a Z
- letters or digits may NOT be repeated
- the code ends with an even digit.

Asked: 2024

12.5.3 The company wishes to increase the number of codes in QUESTION 12.5.2 by allowing the letters B, C, D and F to be used. Calculate the percentage increase in the number of different codes that can now be formed. (2)

Asked: 2024

A scooter company makes scooters with six different body styles. Each scooter has one seat colour and one body colour, chosen as shown in the table below. The seat colour of a scooter must NOT be the same as its body colour.

BODY STYLES	6 different body styles
SEAT COLOURS	Red ; Black ; Tan ; Grey
BODY COLOURS	Red ; Black ; White ; Blue ; Green

12.6 The company wants to display one of each possible variation of its scooter in a showroom with a floor space of 600 m^2. Each scooter needs a floor space of 5 m^2. Is this display possible? Justify your answer with the necessary calculations. (6)

Asked: 2019 (there the display did not fit)

A teacher has 3 different mathematics books, 4 different science books and 3 different language books. She must arrange these 10 books from left to right on a shelf.

12.7.1 Write down the probability that a language book will be the first book placed on the shelf. (1)

Asked: 2021

12.7.2 Calculate the number of different ways the 10 books can be placed on the shelf if any book can be placed in any position. (2)

Asked: 2021

12.7.3 Calculate the probability that a mathematics book is placed in the first position, the three language books are placed next to each other, and a science book is placed in the last position. (4)

Asked: 2021

A four-digit code is needed to open a safe. The code must be an odd number and may not contain any of the digits 0, 1 or 9. Digits may not be repeated.

12.8.1 How many possible 4-digit codes are there to open the safe? (3)

Asked: 2022

12.8.2 Calculate the probability that you will open the safe at the first attempt if it is given that the code is greater than 5 000 and the second digit is 4. (5)

Asked: 2022

Nine learners stand randomly in a line, one behind the other.

12.9.1 In how many different ways can the nine learners stand in the line? (1)

Asked: 2023

12.9.2 Calculate the probability that there will be 4 learners between the 2 oldest learners in the line. (4)

Asked: 2023

Seven runners take part in a race. There are no ties.

12.10.1 Calculate the total number of possible ways in which the 7 runners can finish the race if Chipo finishes immediately after Dumi. (2)

Asked: 2025

12.10.2 Calculate the probability that TWO OR MORE runners finish the race after Dumi and before Chipo. (4)

Asked: 2025

[65]